

EE6204 Systems Analysis

Lecture 1: Linear Programming

presented by

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Optimization Models

- **Optimization Models**

Optimization models attempt to express, in mathematical terms, the goal of solving a problem in the “best” way. That might mean running a business to maximize profit, minimize loss, maximize efficiency, or minimize risk. It might mean designing a bridge to minimize weight or maximum strength. It might mean selecting a flight plan for an aircraft to minimize time or fuel use. The desire to solve a problem in an optimal way is so common that optimization models arise in almost every area of application. We will look at the following optimization models.

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Linear Programming (LP)

- **Linear Programming (LP)**

A LP model involves the optimization of a linear function subject to linear constraints on the variables.

Example 1

A tailor has the following materials available: 16 m² cotton, 11 m² silk, 15 m² wool. A suit requires the following: 2 m² cotton, 1 m² silk, 1 m² wool. A gown requires the following: 1 m² cotton, 2 m² silk, 3 m² wool. If a suit sells for \$30 and a gown for \$50, how many of each garment should the tailor make to obtain the maximum amount of money?



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Linear Programming (LP)

The following table summarized the above facts:

	Suit	Gown	Available
Cotton	2	1	16
Silk	1	2	11
Wool	1	3	15
Price	\$30	\$50	

The above problem is an optimization problem in which one seeks to maximize a specific quantity, called the objective. The objective depends on a finite number of input variables, which may be related through one or more constraints.



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Linear Programming (LP)

Problem Formulation

Optimization problems most often are stated verbally. The solution procedure is to model the problem with a mathematical program and then solve the program by some techniques.

The following approach can be used to transform a word problem into a mathematical program:

1. Determine the quantity to be optimized and express it as a mathematical function. Doing so serves to define the input variables.
2. Identify all the stipulated restrictions and express them mathematically. These requirements constitute the constraints.



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Linear Programming (LP)

3. Express any hidden conditions. Such conditions are not stipulated explicitly in the problem but are apparent from the physical situation being modeled. Generally, they involve non-negativity or integer requirements on the input variables.

We now apply the steps to the previous problem.

The objective is to maximize the amount of money, denoted by f .

Let x be the number of suits the tailor makes and

y the number of gowns.

We express the objective as

$$\text{Maximize: } f = 30x + 50y \quad (1)$$

The tailor is subject to a cotton constraint. 2 m² of cotton is required for each suit and 1 m² for each gown, but only 16 m² is available. Hence the cotton constraint is

$$2x + y \leq 16 \quad (2)$$



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Linear Programming (LP)

The tailor is also subject to a silk constraint. 1 m² of silk is required for each suit and 2 m² for each gown, but only 11 m² is available. The silk constraint is

$$x + 2y \leq 11 \quad (3)$$

The tailor is also subject to a wool constraint. 1 m² of wool is required for each suit and 3 m² for each gown, but only 15 m² is available. The wool constraint is

$$x + 3y \leq 15 \quad (4)$$

It is obvious that negative quantities of suits or gowns cannot be produced, so the two hidden constraints are $x \geq 0$, $y \geq 0$.

Thus, the mathematical problem is

Maximize: $f = 30x + 50y$

Subject to: $2x + y \leq 16$

$$x + 2y \leq 11$$

$$x + 3y \leq 15$$

with: all variables being nonnegative and integer.



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Linear Programming (LP)

Example 2

A person requires 10, 12 and 12 units of chemicals A, B and C, respectively, for his garden. A liquid product contains 5, 2, 1 units of A, B and C, respectively per jar; and a dry product contains 1, 2 and 4 units of A, B, and C, respectively, per carton. If the liquid product sells for \$3 per jar and the dry product sells for \$2 per carton, how many of each should he purchase to minimize the cost and meet the requirements?

The following table summarized the above facts:

	Units per jar	Units per Carton	Units needed
Chem A	5	1	10
Chem B	2	2	12
Chem C	1	4	12
Cost	\$3	\$2	



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Linear Programming (LP)

Let x denote the number of jars bought and
 y the number of cartons.

We seek to minimize the function

$$F = 3x + 2y$$

Subject to the conditions

$$5x + y \geq 10$$

$$2x + 2y \geq 12$$

$$x + 4y \geq 12$$

And the fact that x and y are nonnegative:

$$x \geq 0, y \geq 0$$



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Linear Programming (LP)

• Linear Programming Problems

LP problems are concerned with finding the maximum or minimum of a linear function

$$Z = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

Subject to linear constraints on the variables

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

.....

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

Where all $x_j \geq 0$ and c_j , b_i and a_{ij} are known constants.

Any other mathematical program is nonlinear. The following is a nonlinear program.

$$\text{Minimize: } f = x_1^2 + x_2^2$$

$$\text{Subject to: } x_1 - x_2 = 3, x_2 \geq 2$$



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Linear Programming (LP)

- Terminology for Solutions

In any mathematical program, we seek a solution. Any specification of values for the input variables $(x_1, x_2, \dots, x_n)$ is called a solution, regardless of whether it is a desirable or even an allowable choice. Different types of solutions are then identified by using an appropriate adjective.

A feasible solution is a solution for which all the constraints are satisfied.

The feasible region is the collection of all feasible solutions.

It is possible for a problem to have no feasible solutions.

An infeasible solution is a solution for which at least one constraint is violated.



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Linear Programming (LP)

Given that there are feasible solutions, the goal of LP is to find a best feasible solution, as measured by the value of the objective function.

An optimal solution is a feasible solution that has the most favorable value of the objective function.

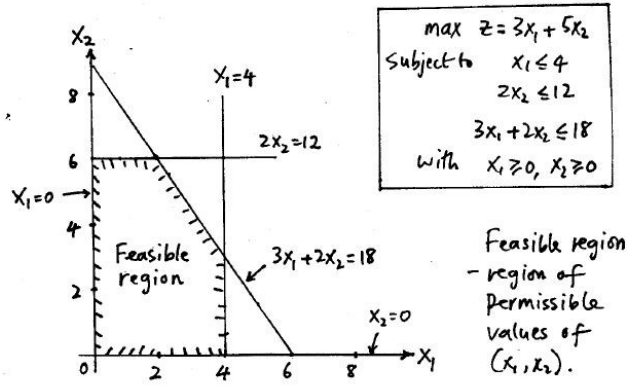
If the objective function is to be maximized (minimized), then the most favorable value is the largest value (smallest).



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Linear Programming (LP)

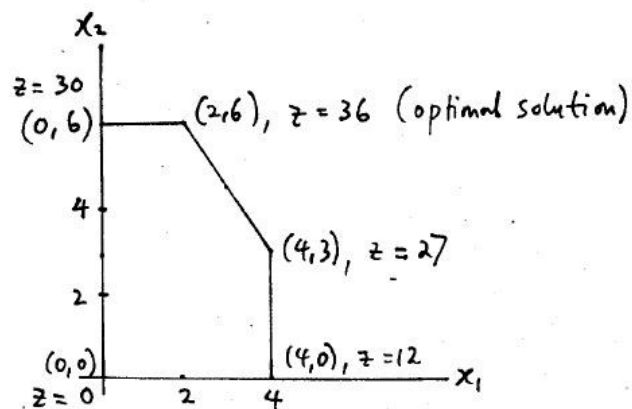


Points (2, 3) and (4, 1) are feasible solutions.
Points (-1, 3) and (4, 4) are infeasible solutions.



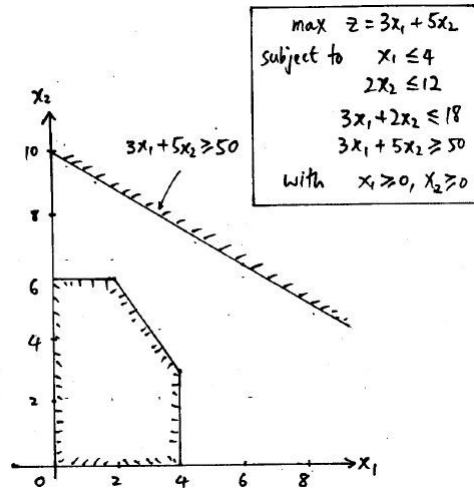
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Linear Programming (LP)



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Linear Programming (LP)



Linear Programming (LP)

• LP: Standard Form

The simplex method for solving linear programs involving many variables will be covered later. To initialize the method, one must transform all *inequality* constraints into *equalities* and must know one feasible, nonnegative solution.

Non-negativity conditions

All variables are constrained to be nonnegative. Linear constraints are of the form

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n \sim b_i$$

Where $\sim$ stands for $=$, $\geq$ or $\leq$. The constants b_i may always be assumed nonnegative.

Example

The constraint $3x_1 - 4x_2 + 5x_3 \leq -6$ is multiplied by -1 to obtain $-3x_1 + 4x_2 - 5x_3 \geq 6$.

Linear Programming (LP)

Slack variables

A linear constraint of the form

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n < b_i \quad (1)$$

can be converted into an equality by adding a new, nonnegative variable to the left-hand side of the inequality, x_{n+i} where

$$x_{n+i} = b_i - (a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n) \geq 0$$

Such a variable is known as a slack variable. Thus, the constraint (1) becomes

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n + x_{n+i} = b_i$$

Example

The constraint $3x_1 + 5x_2 + 2x_3 + 4x_4 \leq 100$ becomes $3x_1 + 5x_2 + 2x_3 + 4x_4 + x_5 = 100$, where x_5 is the slack variable.



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Linear Programming (LP)

Surplus variables

A linear constraint of the form

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n > b_i \quad (2)$$

can be converted into an equality by subtracting a new, nonnegative variable from the left-hand side of the inequality, x_{n+i} where

$$x_{n+i} = a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n - b_i \geq 0$$

Such a variable is known as a surplus variable. Thus, the constraint (2) becomes

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n - x_{n+i} = b_i$$

Example

The constraint $3x_1 + 5x_2 + x_3 \geq 40$ becomes $3x_1 + 5x_2 + x_3 - x_4 = 40$, where x_4 is the surplus variable.



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Linear Programming (LP)

Similarly, an unrestricted variable (i.e., a variable without lower or upper bounds) may be replaced by a difference of 2 new nonnegative variables. For example, if x_2 is an unrestricted variable, then throughout the problem it will be replaced by $x_2 = x_2' - x_2''$.



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Linear Programming (LP)

Consider the example:

$$\min: z = 5x_1 + 3x_2 - 7x_3$$

$$\text{s.t.: } 2x_1 + 4x_2 + 6x_3 = 7$$

$$3x_1 - 5x_2 + 3x_3 \leq 5$$

$$-4x_1 - 9x_2 + 4x_3 \leq -4$$

With: $x_1 \geq -2$, $0 \leq x_2 \leq 4$, x_3 free

The third constraint is multiplied by -1 so that all the right-hand side of the constraints are nonnegative:

$$4x_1 + 9x_2 - 4x_3 \geq 4$$

The variable x_3 is transformed to $x_3 = x_3' - x_3''$ because it is a free variable. Variable x_1 is transformed to $x_1' = x_1 + 2$. The upper bound $x_2 \leq 4$ will be treated as one of the general constraints.



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Linear Programming (LP)

With these substitutions we obtain

$$\min: z = 5x_1' + 3x_2 - 7x_3' + 7x_3'' - 10$$

$$\text{s.t.: } 2x_1' + 4x_2 + 6x_3' - 6x_3'' = 11$$

$$3x_1' - 5x_2 + 3x_3' - 3x_3'' \leq 11$$

$$4x_1' + 9x_2 - 4x_3' + 4x_3'' \geq 4$$

$$x_2 \leq 4$$

With: $x_1', x_2, x_3', x_3'' \geq 0$

The constant term in the objective, “-10” is usually removed via a transformation of the form $z' = z + 10$ so that the revised objective is

$$\text{Min: } z' = 5x_1' + 3x_2 - 7x_3' + 7x_3''$$

The final step in the conversion is to add slack and surplus variable to convert the general constraints to equalities:



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Linear Programming (LP)

$$\min: z' = 5x_1' + 3x_2 - 7x_3' + 7x_3''$$

$$\text{s.t.: } 2x_1' + 4x_2 + 6x_3' - 6x_3'' = 11$$

$$3x_1' - 5x_2 + 3x_3' - 3x_3'' + x_4 = 11$$

$$4x_1' + 9x_2 - 4x_3' + 4x_3'' - x_5 = 4$$

$$x_2 + x_6 = 4$$

With: $x_1', x_2, x_3', x_3'', x_4, x_5, x_6 \geq 0$

x_4 and x_6 are the slack variables and x_5 is the surplus variable.



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Linear Programming (LP)

- **Generating an initial feasible solution**

After all linear constraints (with nonnegative right-hand sides) have been transformed into equalities by introducing slack and surplus variables where necessary, *add a new variable called an artificial variable, to the left-hand side of each constraint equation that does not contain a slack variable.*

Each constraint equation will then contain either one slack variable or one artificial variable.

A nonnegative initial condition to this new set of constraints is obtained by setting each slack variable and artificial variable equal to the right-hand side of the equation in which it appears and setting all other variables, including the surplus variables, equal to zero.



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Linear Programming (LP)

- **Generating an initial feasible solution**

The set of constraints

$$x_1 + 2x_2 \leq 3$$

$$4x_1 + 5x_2 \geq 6$$

$$7x_1 + 8x_2 = 15$$

is transformed to

$$x_1 + 2x_2 + x_3 = 3$$

$$4x_1 + 5x_2 - x_4 + \bar{x}_5 = 6$$

$$7x_1 + 8x_2 + \bar{x}_6 = 15$$

where

x_3 – slack variable, x_4 – surplus variable, $\bar{x}_5, \bar{x}_6$ – artificial variable

A nonnegative solution to the new system is:

$x_3 = 3, \bar{x}_5 = 6, \bar{x}_6 = 15$, and $x_1 = x_2 = x_4 = 0$.

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Linear Programming (LP)

- On the use of artificial variables

It is necessary to find an *initial basic feasible solution* to start the simplex method.

Consider the original program:

$$\min: z = 3x_1 + 5x_2$$

$$\text{Subject to: } x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

$$\text{with: } x_1 \geq 0, x_2 \geq 0$$

In standard form we have

$$\min: z = 3x_1 + 5x_2$$

$$\text{Subject to: } x_1 + x_3 = 4$$

$$2x_2 + x_4 = 12$$

$$3x_1 + 2x_2 + x_5 = 18$$

with: all variables nonnegative.



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Linear Programming (LP)

There are 5 variables and 3 constraints.

$\Rightarrow$ 2 degree of freedom. We can set $x_1 = x_2 = 0$ (nonbasic variables) and solve for the basic variables x_3, x_4, x_5 .

Then, by inspection

$$(x_3, x_4, x_5)^T = (4, 12, 18)^T$$

No work required to solve for the basic variables!

$(0, 0, 4, 12, 18)$ is the initial basic feasible solution.

$$(x_1, x_2, x_3, x_4, x_5)$$



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Linear Programming (LP)

Consider the following program

$$\text{min: } z = 2x_1 + 3x_2$$

$$\text{subject to: } 3x_1 + 2x_2 = 14$$

$$2x_1 - 4x_2 \geq 12$$

$$4x_1 + 3x_2 \leq 19$$

$$\text{with: } x_1 \geq 0, x_2 \geq 0$$

In standard form, we have

$$\text{min: } z = 2x_1 + 3x_2$$

$$\text{subject to: } 3x_1 + 2x_2 = 14 \quad (1)$$

$$2x_1 - 4x_2 - e = 12 \quad (2)$$

$$4x_1 + 3x_2 + s = 19 \quad (3)$$

with: all variables nonnegative.

There are 4 variables and 3 constraints $\Rightarrow$ only 1 degree of freedom.



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Linear Programming (LP)

These equations do not have obvious initial basic feasible solution, because there is no longer slack variables to use as the initial basic variables for eqn (1) and (2).

This difficulty can be overcome by introducing artificial variables into eqn (1) and (2) as they do not contain slack variables. This gives:

$$3x_1 + 2x_2 + a_3 = 14$$

$$2x_1 - 4x_2 - e_4 + a_5 = 12$$

$$4x_1 + 3x_2 + s_6 = 19$$

with all variables nonnegative.

There are now 6 variables and 3 constraints $\Rightarrow$ 3 degrees of freedom.

Set $(x_1, x_2, e_4)^T = (0, 0, 0)^T$.



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Linear Programming (LP)

This gives, with no work required,

$$(a_3, a_5, s_6)^T = (14, 12, 19)^T$$

- ❖ The purpose of introducing the artificial variables is to get an initial basic feasible solution to start the simplex method.
- ❖ However, the artificial variables are not part of the original program. They are extra variables that are temporarily added to the problem.



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Linear Programming (LP)

• Penalty costs

The introduction of the slack and surplus variables alters neither the nature of the constraints nor the objective. Accordingly, such variables are incorporated into the objective function with zero coefficients.

Artificial variables, however, do change the nature of the constraints. Since they are added to only one side of an equality, the new equality, the new system is equivalent to the old system of constraints if and only if the artificial variables are zero.

To guarantee such assignments in the optimal solution, artificial variables are incorporated into the objective function with very large positive (negative) coefficients in a minimization (maximization) program.



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Linear Programming (LP)

Consider the previous example.

If now artificial variables $\bar{x}_7$ and $\bar{x}_8$ are added to the first and third constraints, the result is

$$\min: z' = 5x_1' + 3x_2 - 7x_3' + 7x_3'' + 0x_4 + 0x_5 + 0x_6 + M\bar{x}_7 + M\bar{x}_8$$

(M is a "very large" number)

subject to:

$$2x_1' + 4x_2 + 6x_3' - 6x_3'' + \bar{x}_7 = 11$$

$$3x_1 - 5x_2 + 3x_3' - 3x_3'' + x_4 = 11$$

$$4x_1' + 9x_2 - 4x_3' + 4x_3'' - x_5 + \bar{x}_8 = 12$$

$$x_2 + x_6 = 4$$

$$\text{with: } x_1', x_2, x_3', x_3'', x_4, x_5, x_6, \bar{x}_7, \bar{x}_8 \geq 0$$

Initial feasible solution is $x_4 = 11, x_6 = 4, \bar{x}_7 = 11, \bar{x}_8 = 12$, and $x_1' = x_2 = x_3' = x_3'' = x_5 = 0$



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Linear Programming (LP)

• Standard form

A linear program is in standard form if the constraints are all modelled as equalities and if one feasible solution is known. In matrix notation, the standard form is

$$\text{optimize: } z = C^T X$$

$$\text{subject to: } AX = B$$

$$\text{with: } X \geq 0$$

where X is the column vector of unknowns, including all slack, surplus and artificial variables; C^T is the row vector of the corresponding costs; A is the coefficient matrix of the constraint equations; and B is the column vector of the right-hand sides of the constraint equations.



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Linear Programming (LP)

If X_0 denotes the vector of slack and artificial variables only, then the initial feasible solution is given by $X_0 = B$ where it is understood that all variables in X not included in X_0 are assigned zero values.

- **Standard Matrix form**

Put the following program in standard matrix form;

$$\min: z = 6x_1 + 3x_2 + 4x_3$$

$$\text{subject to: } x_1 + 6x_2 + x_3 = 10$$

$$2x_1 + 3x_2 + x_3 = 15$$

with: all variables nonnegative



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Linear Programming (LP)

The program in standard form is

$$\min: z = 6x_1 + 3x_2 + 4x_3 + M\bar{x}_4 + M\bar{x}_5$$

$$\text{subject to: } x_1 + 6x_2 + x_3 + \bar{x}_4 = 10$$

$$2x_1 + 3x_2 + x_3 + \bar{x}_5 = 15$$

(1)

with: all variables nonnegative.

An initial feasible solution is $\bar{x}_4 = 10$, $\bar{x}_5 = 15$, and $x_1 = x_2 = x_3 = 0$.

System (1) is in standard matrix form if we define:

$$X = [x_1 \ x_2 \ x_3 \ \bar{x}_4 \ \bar{x}_5]^T$$

$$C = [6 \ 3 \ 4 \ M \ M]^T$$

$$A = \begin{bmatrix} 1 & 6 & 1 & 1 & 0 \\ 2 & 3 & 1 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 10 \\ 15 \end{bmatrix}$$

$$X_0 = [\bar{x}_4 \ \bar{x}_5]^T$$



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Linear Programming (LP)

Put the following program in standard matrix form:

$$\max: z = x_1 + x_2$$

$$\text{subject to: } x_1 + 5x_2 \leq 5$$

$$2x_1 + x_2 \leq 4$$

$$\text{with: } x_1, x_2 \geq 0$$



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Linear Programming (LP)

The program in standard form is

$$\max: z = x_1 + x_2 + 0x_3 + 0x_4$$

$$\text{subject to: } \left. \begin{array}{l} x_1 + 5x_2 + x_3 = 5 \\ 2x_1 + x_2 + x_4 = 4 \end{array} \right\} \quad (1)$$

with: all variables nonnegative.

An initial feasible solution is $x_3 = 5$, $x_4 = 4$ and $x_1 = x_2 = 0$.

System (1) is in standard matrix form if we define:

$$X = [x_1 \ x_2 \ x_3 \ x_4]^T$$

$$C = [1 \ 1 \ 0 \ 0]^T$$

$$A = \begin{bmatrix} 1 & 5 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

$$X_0 = [x_3 \ x_4]^T$$



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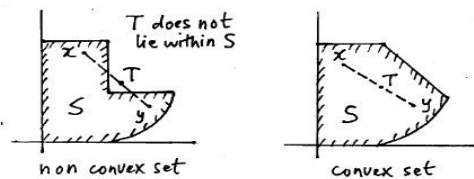
Linear Programming (LP)

• Solutions

Convex set

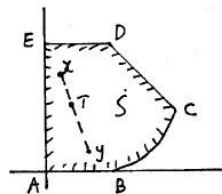
A set S is convex, if, for any elements x and y of S , $\alpha x + (1 - \alpha)y$ is an element of S for $0 \leq \alpha \leq 1$.

A convex set is a collection of points such that for each pair of points in the collection, the entire line segment joining these two points is also in the collection.



Linear Programming (LP)

An extreme point of a convex set is a point that does not lie on any line segment that joins two other points in the set.



Points A, B, C, D, E and all points on the boundary between B and C are extreme points.

Linear Programming (LP)

Extreme point solutions

Let F designate the set of all feasible solutions to the linear program in standard form; that is, F is the set of all vectors X that satisfy $AX = B$ and $X \geq 0$.

F is a convex set having a finite number of extreme points.

The objective function attains its optimum at an extreme point of F , provided an optimum exists.

Basic feasible solutions

Denote the columns of the $m \times n$ matrix in the standard form by $A_1, A_2, \dots, A_n$, respectively. Then the matrix constraint equation $AX = B$ can be rewritten in the vector form

$$x_1 A_1 + x_2 A_2 + \dots + x_n A_n = B \quad (1)$$

Where the A -vectors and B are known m -dimension vectors.



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Linear Programming (LP)

We wish to find nonnegative solutions for the variables $x_1, x_2, \dots, x_n$. Suppose that $m \leq n$ and $\text{rank}(A) = m$, which means that at least a set of m A -vectors is linearly independent.

A basic solution to (1) is obtained by setting $(n-m)$ of the x -variables equal to zero and finding a solution for the remaining m x -variables, provided the m A -vectors corresponding to the x -variables not set to zero are linearly independent. There are nC_m systems of m equations in m unknowns to solve. However, only nonnegative solutions are selected.

The m x -variables not initially set equal to zero are called basic variables.

A point x is a basic feasible solution, if, in addition basic solution, satisfies the non-negativity constraint $x \geq 0$.

The objective function attains its optimum at a basic feasible solution.



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Linear Programming (LP)

The extreme points of F are precisely the basic feasible solutions.
We seek among the basic feasible solutions the one(s) at which the objective is optimized.

Note

If one or more of the basic variables turns out to be zero, the basic feasible solution is *degenerate*. From the practical standpoint, the condition reveals that the model has at least one redundant constraint. If all the basic variables are positive, the basic feasible solution is *nondegenerate*.



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Linear Programming (LP)

- **Basic feasible solutions**

Consider the example Ex 2.

$$\text{min: } z = -2x_1 - x_2$$

$$\text{subject to: } 3x_1 + 4x_2 \leq 12$$

$$2x_1 + x_2 \leq 6$$

$$\text{with: } x_1, x_2 \geq 0$$



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Linear Programming (LP)

Reformulate the problem as

$$\begin{aligned} \min: z &= -2x_1 - x_2 + 0x_3 + 0x_4 \\ \text{subject to: } & \left. \begin{aligned} 3x_1 + 4x_2 + x_3 &= 12 \\ 2x_1 + x_2 + x_4 &= 6 \end{aligned} \right\} \quad (1) \end{aligned}$$

with: all variables nonnegative.

Since $n - m = 4 - 2 = 2$, a basic solution will have 2 x-variable set equal to 0.

There are ${}^nC_m = {}^4C_2 = 6$ systems of 2 equations in 2 unknowns to solve.

System (1) can be written as

$$x_1 \begin{pmatrix} 3 \\ 2 \end{pmatrix} + x_2 \begin{pmatrix} 4 \\ 1 \end{pmatrix} + x_3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 12 \\ 6 \end{pmatrix}$$

$A_1 \quad A_2 \quad A_3 \quad A_4$



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Linear Programming (LP)

With $x_1 = x_2 = 0$, the vector constraint becomes

$$x_3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 12 \\ 6 \end{pmatrix}$$

Which gives $x_3 = 12$, $x_4 = 6$. Since A_3 and A_4 are linearly independent (A_3 and A_4 are columns of the 2×2 identity matrix), the complete solution* $(0 \ 0 \ 12 \ 6)^T$ is basic. Here, the basic variables x_3, x_4 are > 0 .

* Note that this is the initial solution x_0 generated in the previous section, and is a basic feasible solution.



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Linear Programming (LP)

With $x_2 = x_4 = 0$, the vector constraint becomes

$$x_1 \begin{pmatrix} 3 \\ 2 \end{pmatrix} + x_3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 12 \\ 6 \end{pmatrix}$$

which gives $x_1 = 3$, $x_3 = 3$. Since A_1 and A_3 are linearly independent, the complete solution $(3 \ 0 \ 3 \ 0)^T$ is basic.

Thus, the 6 basic solutions are given by

$$\underbrace{\begin{pmatrix} 0 \\ 0 \\ 12 \\ 6 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 12/5 \\ 6/5 \\ 0 \\ 0 \end{pmatrix}}_{\text{basic feasible solutions}}, \underbrace{\begin{pmatrix} 4 \\ 0 \\ 0 \\ -2 \end{pmatrix}, \begin{pmatrix} 0 \\ 6 \\ -12 \\ 0 \end{pmatrix}}_{\text{unfeasible solutions}}$$

basic feasible solutions unfeasible solutions

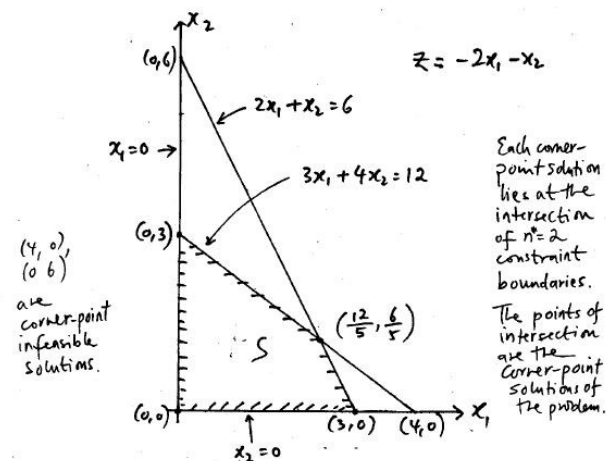
The solution set is shown in the following figure.



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Linear Programming (LP)



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Linear Programming (LP)

There are 4 extreme points in S

$$\begin{array}{l} \text{Corner-} \\ \text{point} \\ \text{feasible} \\ \text{solutions} \end{array} \left\{ \begin{array}{l} (0 \ 0)^T, z = -2(0) - (0) = 0 \\ (0 \ 3)^T, z = -2(0) - (3) = -3 \\ (3 \ 0)^T, z = -2(3) - (0) = -6 \\ (12/5 \ 6/5)^T, z = -2(12/5) - (6/5) = -6 \end{array} \right.$$

Z has a minimum value of -6 at all points of the form

$$X = \alpha \begin{pmatrix} 12/5 \\ 6/5 \end{pmatrix} + (1 - \alpha) \begin{pmatrix} 3 \\ 0 \end{pmatrix}, 0 \leq \alpha \leq 1.$$



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Linear Programming (LP)

In principle, we could enumerate all basic feasible solutions to an LP and find the basic feasible solutions with the optimum value. The problem with this approach is that even with small LPs have a very large number of basic feasible solutions. For example, an LP in standard form that has 20 variables and 10 constraints might have (if each basic solution were feasible) up to ${}^{20}C_{10} = 184,756$ basic feasible solutions.

This process is very time consuming, even for a high speed computer. Using the simplex method will greatly reduce the amount of time and work required to solve the LP problem.



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EE6204 Systems Analysis

Lecture 2: LP: The Simplex Method

presented by

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Linear Programming

- **LP: The Simplex Method**

The simplex tableau

The simplex method is a matrix procedure for solving linear programs in the standard form

Optimize: $z = C^T X$

Subject to: $AX = B$

With: $X \geq 0$

where $B \geq 0$ and a basic feasible solution X_0 is known. * Starting with X_0 , the method locates successively other basic feasible solutions having better values of z , until the optimal solution is obtained.

* The initial solution X_0 generated in the previous section is a basic feasible solution.

2

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Linear Programming

For minimization programs, the simplex method utilized Tableau 1 as shown.

	x^T	
x_0	A	B
	$C^T - C_0^T A$	$-C_0^T B$

C_0 designates the cost vector associated with the variables in X_0 .



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Linear Programming

For maximization programs, the above tableau applies if the elements of the bottom row have their signs reversed (see below).

	x^T	
x_0	A	B
	$-C^T + C_0^T A$	$C_0^T B$

* If we have a maximization program and we want to convert it to a minimization program, we multiply the objective z by -1. After the program has been solved, the optimal objective value must be multiplied by -1.



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Linear Programming

The Simplex method

1. Locate the most negative number in the bottom row of the simplex tableau, excluding the last column, and call the column in which this number appears the work column. If more than one candidate for most negative number exists, choose one.
2. Pick out each coefficient (> 0) in the work column excluding the last row, divide each of these coefficients into the right side entry for the same row. Designate the element in the work column that yields the smallest ratio as the pivot element. If more than one element yields the same smallest ratio, choose one. If no element in the work column is > 0 , the program has no solution.



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Linear Programming

3. Use elementary row operations * to convert the pivot element to 1 and then reduce all other elements in the work column to 0.
4. Replace the x-variable in the pivot row and column zero by the x-variable in the row zero and pivot column. The new column zero is the current set of basic variables.
5. Repeat steps 1 through 4 until there are no negative numbers in the last row, excluding the last column.
6. The optimal solution is obtained by assigning to each variable in column zero that value in the corresponding row and last column. All other values are assigned the value 0. The associated z, the optimal value of the objective function, is the number in the last row and last column for a maximization program, but the negative of this number for a minimization program.



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Linear Programming

* Elementary row operations

- Multiply or divide a row by a nonzero constant; add or subtract a multiple of one row to another row.



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Linear Programming

• **LP: The Simplex Method**

Solve the following program:

$$\text{Min: } z = -2x_1 - x_2$$

Subject to:

$$3x_1 + 4x_2 \leq 12$$

$$2x_1 + x_2 \leq 6$$

$$\text{With: } x_1, x_2 \geq 0$$



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Linear Programming

Reformulate the problem as

$$\text{Min: } z = -2x_1 - x_2 + 0x_3 + 0x_4$$

Subject to:

$$3x_1 + 4x_2 + x_3 = 12$$

$$2x_1 + x_2 + x_4 = 6$$

With: all variables nonnegative.

The simplex tableau is

	X^T	
X_0	A	B
	$C^T - C_0^T A$	$-C_0^T B$

where

$$x = [x_1 \ x_2 \ x_3 \ x_4]^T$$



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Linear Programming

$$C^T = [-2 \ -1 \ 0 \ 0]$$

$$A = \begin{bmatrix} 3 & 4 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 12 \\ 6 \end{bmatrix}$$

$$C_0^T = [0 \ 0],$$

$$X_0 = [x_3 \ x_4]^T$$

and

$$C^T - C_0^T A = [-2 \ -1 \ 0 \ 0]$$

$$C_0^T B = 0$$

Thus, the simplex tableau is



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Linear Programming

	x_1	x_2	x_3	x_4	
x_3	3	4	1	0	12
x_4	2	1	0	1	6
	-2	-1	0	0	0

We are now ready to apply the simplex method. The most negative element in the last row of the tableau above is -2, corresponding to the x_1 – column; hence this column becomes the work column. x_1 is to be changed to a basic variable.

	x_1	x_2	x_3	x_4	
x_3	3	4	1	0	12
x_4	2*	1	0	1	6
	-2	-1	0	0	0

Ratio
 $12/3 = 4$
 $6/2 = 3$

↑
 minimum



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Linear Programming

Forming the ratios $12/3 = 4$ and $6/2 = 3$, we find the element 2, marked by the asterisk * in the above tableau, is the pivot element, since it yields the smallest ratio. Then applying step 3 gives the following tableau

	x_1	x_2	x_3	x_4	
x_3	3	4	1	0	12
x_4	1	1/2	0	1/2	3
	-2	-1	0	0	0

← $\frac{1}{2} R_2$

* x_4 is the leaving basic variable. In the next tableau, x_1 replaces x_4 as the basic variable for row 2. For a basic variable, it must have a coefficient of 1 in this row of the simplex tableau and a coefficient of 0 in every other row, including the last.



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Linear Programming

This is the proper form from *Gaussian elimination*.

Applying step 3 gives

	x_1	x_2	x_3	x_4		
x_3	0	$5/2$	1	$-3/2$	3	$\leftarrow -3R_2 + R_1$
$x_4 \rightarrow x_1$	1	$1/2$	0	$1/2$	3	
	0	0	0	1	6	$\leftarrow 2R_2 + R_3$

nonnegative estimate

Since the last row of the above tableau contains no negative elements, it follows from step 6 that the optimal solution is $x_1^* = 3$, $x_3^* = 3$, $x_2^* = x_4^* = 0$, with $z^* = -6$ (minimization).

* Note that the above program has more than one optimal solution (See Example Ex2).



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Linear Programming

Solve the following program

Max: $z = x_1 + x_2$

Subject to:

$$x_1 + 5x_2 \leq 5$$

$$2x_1 + x_2 \leq 4$$

With: x_1, x_2 nonnegative



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Linear Programming

Reformulate the problem as

$$\text{Max: } z = x_1 + x_2 + 0x_3 + 0x_4$$

Subject to:

$$x_1 + 5x_2 + x_3 = 5$$

$$2x_1 + x_2 + x_4 = 4$$

With: all variables nonnegative

The simplex tableau is

	X^T	
X_0	A	B
	$C_0^T A - C^T$	$C_0^T B$



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Linear Programming

where

$$X = [x_1 \ x_2 \ x_3 \ x_4]^T$$

$$C = [1 \ 1 \ 0 \ 0]^T$$

$$A = \begin{bmatrix} 1 & 5 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

$$C_0 = [0 \ 0]^T,$$

$$X_0 = [x_3 \ x_4]^T$$

$$C_0^T A - C^T = [-1 \ -1 \ 0 \ 0]$$

$$C_0^T B = 0$$



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Linear Programming

The simplex tableau is thus given by

	x_1	x_2	x_3	x_4	
x_3	1	5	1	0	5
x_4	2*	1	0	1	4
	-1	-1	0	0	0

Ratio
 $5/1 = 5$
 $4/2 = 2$



	x_1	x_2	x_3	x_4	
x_3	1	5	1	0	5
x_4	1	1/2	0	1/2	2
	-1	-1	0	0	0

← $1/2R_2$



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Linear Programming

	x_1	x_2	x_3	x_4	
x_3	0	9/2*	1	-1/2	3
$x_4 \rightarrow x_1$	1	1/2	0	1/2	2
	0	-1/2	0	1/2	2

← $-R_2 + R_1$
 $3/(9/2) = 2/3$

← $R_2 + R_3$
 $2/(1/2) = 4$



	x_1	x_2	x_3	x_4	
x_3	0	1	2/9	-1/9	2/3
x_1	1	1/2	0	1/2	2
	0	-1/2	0	1/2	2

← $2/9R_1$



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Linear Programming

	x_1	x_2	x_3	x_4	
$\rightarrow X_3 \rightarrow X_2$	0	1	$2/9$	$-1/9$	$2/3$
x_1	1	0	$-1/9$	$5/9$	$5/3$
	0	0	$1/9$	$4/9$	$7/3$

$\leftarrow -1/2R_1 + R_2$

$\leftarrow 1/2R_1 + R_3$

Thus, the optimal solution is

$$x_1^* = 5/3, x_2^* = 2/3, x_3^* = x_4^* = 0$$

with $z^* = 7/3$ (maximization)



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Linear Programming

- Modifications for programs with artificial variables**

Whenever artificial variables are part of the initial solution X_0 , the last row of Tableau 1 will contain the penalty cost M . The following modifications are incorporated into the simplex method; the resulting algorithm is the two-phase method.

C1. The last row of Tableau 1 is decomposed into two rows, the first of which involves those terms not containing M , while the second involves the coefficients of M in the remaining terms.

Suppose the last row of Tableau 1 is

$$-9-8M \quad 0 \quad -9-9M \quad 0 \quad M \quad 0 \quad -14-2M$$

Under C1 it would be transformed into the two rows

$$-9 \quad 0 \quad -9 \quad 0 \quad 0 \quad 0 \quad -14$$

$$-8 \quad 0 \quad -9 \quad 0 \quad 1 \quad 0 \quad -2$$

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Linear Programming

C2. Step 1 of the simplex method is applied to the last row created in C1 (followed by steps 2, 3, 4), until this row contains no negative elements. Then Step 1 is applied to those elements in the next-to-last row that are positioned over 0's in the last row.

C3. Whenever an artificial variable ceases to be basic (i.e., is removed from column zero of the tableau as a result of Step 4) it is deleted from the top row of the tableau, as is the entire column under it.

C4. The last row can be deleted from the tableau whenever it contains all 0's.

C5. If nonzero artificial variables are present in the final basic set, then the program has no solution.

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Linear Programming

Note:

The two-phase method consists of:

Phase 1, which drives out the artificial variables by finding a basic feasible solution for which the artificial variables have value zero and are nonbasic, and

Phase 2, which starts from this basic feasible solution and produced an optimal solution.



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Linear Programming

Two-phase method

Solve the following program:

$$\text{Min: } z = 80x_1 + 60x_2$$

Subject to:

$$0.2x_1 + 0.32x_2 \leq 0.25$$

$$x_1 + x_2 = 1$$

With: x_1 and x_2 nonnegative



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Linear Programming

The program is converted to standard matrix form, with

$$X = [x_1 \ x_2 \ x_3 \ x_4]^T$$

$$C = [80 \ 60 \ 0 \ M]^T$$

$$A = \begin{bmatrix} 0.2 & 0.32 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 0.25 \\ 1 \end{bmatrix}$$

$$X_0 = [x_3 \ x_4]^T, C_0 = [0 \ M]^T$$

$$C^T - C_0^T A = [80-M \ 60-M \ 0 \ 0]$$

$$C_0^T B = M$$

Thus, the initial tableau for the 2-phase method is



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Linear Programming

	x_1	x_2	x_3	$\bar{x}_4$	
x_3	0.2	0.32	1	0	0.25
$\bar{x}_4$	1	1	0	1	1
	80 - M	60 - M	0	0	-M

	x_1	x_2	x_3	$\bar{x}_4$	
x_3	0.2	0.32	1	0	0.25
$\bar{x}_4$	1*	1	0	1	1
	80	60	0	0	0
	-1	-1	0	0	-1

Ratio
 $0.25/0.2 = 1.25$
 $1/1 = 1$



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Linear Programming

	x_1	x_2	x_3	$\bar{x}_4$	
x_3	0	0.12	1	-0.2	0.05
$\bar{x}_4 \rightarrow x_1$	1	1	0	1	1
	0	-20	0	-80	-80
	0	0	0	1	0

$-0.2R_2 + R_1$

$-80R_2 + R_3$

$R_2 + R_4$

Nonnegative elements

Col 4 removed

	x_1	x_2	x_3	
x_3	0	0.12*	1	0.05
x_1	1	1	0	1
	0	-20	0	-80

Ratio
 $0.05/0.12 = 5/12$
 $1/1 = 1$



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Linear Programming

	x_1	x_2	x_3	
x_3	0	1	8.333	0.4166
x_1	1	1	0	1
	0	-20	0	-80

$R_1/0.12$

	x_1	x_2	x_3	
$x_3 \rightarrow x_2$	0	1	8.333	0.4166
x_1	1	0	-8.333	0.5833
	0	0	166.667	-71.667

$-R_1 + R_2$

$20R_1 + R_3$



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Linear Programming

Thus, the optimal solution is

$$x_1^* = 0.583, x_2^* = 0.4166, x_3^* = \bar{x}_4^* = 0$$

With $z^* = 71.667$ (minimization)



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Linear Programming

Example of an infeasible problem

Consider the linear program

$$\text{Min: } z = 2x_1 + 3x_2$$

Subject to:

$$x_1 + 2x_2 \leq 2$$

$$6x_1 + 4x_2 \geq 24$$

With: all variables nonnegative.

The program is put in standard form by introducing a slack variable x_3 to the first constraint, and both a surplus variable x_4 and an artificial variable x_5 to the second constraint.

The initial tableau for the 2-phase method is:



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Linear Programming

	x_1	x_2	x_3	x_4	$\bar{x}_5$	
x_3	1	2	1	0	0	2
$\bar{x}_5$	6	4	0	-1	1	24
	2 -6M	3 -4M	0	M	0	-24M

	x_1	x_2	x_3	x_4	$\bar{x}_5$	
x_3	1*	2	1	0	0	2
$\bar{x}_5$	6	4	0	-1	1	24
	2	3	0	0	0	0
	-6	-4	0	1	0	-24

Ratio

$$2/1 = 2$$

$$24/6 = 4$$



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Linear Programming

	x_1	x_2	x_3	x_4	$\bar{x}_5$	
$x_3 \rightarrow x_1$	1	2	1	0	0	2
$\bar{x}_5$	0	-8	-6	-1	1	12
	0	-1	-2	0	0	-4
	0	8	6	1	0	-12

Nonnegative elements

$-6R_2 + R_2$
 $-2R_1 + R_3$
 $6R_1 + R_4$

In the next-to-last row there is no negative entry positioned above a zero of the last row. Thus, the two-phase method signals that optimality has been achieved. But the nonzero artificial variable $\bar{x}_5$ is still basic. By C5, the original program has no solution.

Linear Programming

• Duality

Every linear program in the variables $x_1 x_2 \dots x_n$ has associated with it another linear program in the variables $w_1 w_2 \dots w_m$ (m is the number of constraints in the original program) called the dual. The original problem is called the primal.

Thus, if the primal problem has n variables and m constraints, then the dual problem will have m variables and n constraints.

Linear Programming

The *dual* of a *primal* linear program in the nonstandard matrix form

$$\left. \begin{array}{l} \text{Minimize: } z = C^T X \\ \text{Subject to: } AX \geq B \\ \text{With: } X \geq 0 \end{array} \right\} \quad (1)$$

is the linear program

$$\left. \begin{array}{l} \text{Maximize: } z = B^T W \\ \text{Subject to: } A^T W \leq C \\ \text{With: } W \geq 0 \end{array} \right\} \quad (2)$$

The coefficient in the objective of the primal are the coefficients of the right-hand side of the dual, and vice-versa. The constraint matrix in the dual is the transpose of the matrix in the primal.



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Linear Programming

Programs (1) and (2) are symmetrical in that both involve nonnegative variables and inequality constraints; they are known as the symmetric duals of each other. The dual variables $w_1 w_2 \dots w_m$ are sometimes called shadow costs.

Suppose the *primal* is a max problem as given below. Then the dual will be a min problem.

<u>Primal</u>	<u>Dual</u>
max: $z = C^T X$	min: $z = B^T W$
s.t. $AX \leq B$	s.t. $A^T W \geq C$
with: $X \geq 0$	with: $W \geq 0$

Note: The dual of the dual is always the primal.



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Linear Programming

The symmetric dual of the program

$$\text{Min: } z = 5x_1 + 2x_2 + x_3$$

Subject to:

$$2x_1 + 3x_2 + x_3 \geq 20$$

$$6x_1 + 8x_2 + 5x_3 \geq 30$$

$$7x_1 + x_2 + 3x_3 \geq 40$$

$$x_1 + 2x_2 + 4x_3 \geq 50$$

With: all variables nonnegative

is

$$\text{Max: } z = 20w_1 + 30w_2 + 40w_3 + 50w_4$$

Subject to:

$$2w_1 + 6w_2 + 7w_3 + w_4 \leq 5$$

$$3w_1 + 8w_2 + w_3 + 2w_4 \leq 2$$

$$w_1 + 5w_2 + 3w_3 + 4w_4 \leq 1$$

With: all variables nonnegative

w_j is the dual variable
corresponding to the j th
primal constraint

The i th dual constraint
corresponding to the
primal variable x_i .



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Linear Programming

The symmetric dual of the program

$$\text{Max: } z = 2x_1 + x_2$$

Subject to:

$$x_1 + 5x_2 \leq 10$$

$$x_1 + 3x_2 \leq 6$$

$$2x_1 + 2x_2 \leq 8$$

With: all variables nonnegative

is

$$\text{Min: } z = 10w_1 + 6w_2 + 8w_3$$

Subject to:

$$w_1 + w_2 + 2w_3 \geq 2$$

$$5w_1 + 3w_2 + 2w_3 \geq 1$$

With: all variables nonnegative



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Linear Programming

- **Dual Solutions**

Duality Theorem

If an optimal solution exists either to the primal or symmetric dual program, then the other program also has an optimal solution and the two objective functions have the same optimal value.

In such situations, the optimal solution to the primal (dual) is found in the last row of the simplex tableau for the dual (primal), in those columns associated with the slack or surplus variables. Thus, it may be computationally advantageous to solve a program's dual rather than the program itself.



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Linear Programming

Consider the following program

$$\text{Min: } z = 10x_1 + 6x_2 + 8x_3$$

Subject to:

$$x_1 + x_2 + 2x_3 \geq 2$$

$$5x_1 + 3x_2 + 2x_3 \geq 1$$

With: all variables nonnegative.

To solve this program directly would require the introduction of 2 surplus variables and 2 artificial variables, and the application of the 2-phase method.



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Linear Programming

A simpler approach is to consider its dual program:

Max: $z = 2w_1 + w_2$

Subject to:

$$w_1 + 5w_2 \leq 10$$

$$w_1 + 3w_2 \leq 6$$

$$2w_1 + 2w_2 \leq 8$$

with: all variables nonnegative



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Linear Programming

Reformulate the program as

Max: $z = 2w_1 + w_2 + 0s_1 + 0s_2 + 0s_3$

Subject to:

$$w_1 + 5w_2 + s_1 = 10$$

$$w_1 + 3w_2 + s_2 = 6$$

$$2w_1 + 2w_2 + s_3 = 8$$

With: all variables nonnegative.

The simplex tableau is

	X^T	
X_0	A	B
	$C_0^T A - C^T$	$C_0^T A$



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Linear Programming

	w_1	w_2	s_1	s_2	s_3	
s_1	1	5	1	0	0	10
s_2	1	3	0	1	0	6
s_3	2*	2	0	0	1	8
	-2	-1	0	0	0	0

Ratio
 $10/1 = 10$
 $6/1 = 6$
 $8/2 = 4$

↑

	w_1	w_2	s_1	s_2	s_3	
s_1	1	5	1	0	0	10
s_2	1	3	0	1	0	6
s_3	1	1	0	0	0.5	4
	-2	-1	0	0	0	0

$1/2R_3$



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Linear Programming

Slack variables

	w_1	w_2	s_1	s_2	s_3	
s_1	0	4	1	0	-0.5	6
s_2	0	2	0	1	-0.5	2
w_1	1	1	0	0	0.5	4
	0	1	0	0	1	8

$-R_3 + R_1$

$-R_3 + R_2$

$2R_3 + R_4$

Solution to the primal

The solution to the dual is $w_1^* = 4$, $w_2^* = 0$, with $z^* = 8$.

The solution to the primal is found in the last row of the tableau, in those columns associated with the slack variables.

That is, $x_1^* = x_2^* = 0$, $x_3^* = 1$ with $z^* = 8$.



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Linear Programming

Complementary slackness property

The theorem of complementary slackness is an important result that relates the optimal and dual solutions.

Consider the following *primal* program and its dual

Primal	Dual
min: $z = C^T X$	max: $z = B^T W$
s.t.: $AX \geq B$	s.t.: $A^T W \leq C$
with: $X \geq 0$	with: $W \geq 0$

Let $X = [x_1, \dots, x_n]^T$ and $W = [w_1, \dots, w_m]^T$ be the feasible primal and dual solutions, respectively. Then X is primal optimal and W is dual optimal if and only if

$$e_i^* w_i^* = 0 \quad (i = 1, 2, \dots, m) \quad (a)$$

$$s_j^* x_j^* = 0 \quad (j = 1, 2, \dots, n) \quad (b)$$

where s_i are the slack variables for the dual and e_j are the surplus variables for the primal.



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Linear Programming

Note that there is an interdependency between the *nonnegativity constraints* in one program and the *constraints* in the other program. In (a) (or (b)), if either term is positive, the other must be zero.

Similarly, for the following primal and its dual

Primal	Dual
max: $z = C^T X$	min: $z = B^T W$
s.t.: $AX \leq B$	s.t.: $A^T W \geq C$
with: $X \geq 0$	with: $W \geq 0$

we have

$$s_i^* w_i^* = 0 \quad (i = 1, 2, \dots, m) \quad (a')$$

$$e_j^* x_j^* = 0 \quad (j = 1, 2, \dots, n) \quad (b')$$

where s_i are the slack variables for the primal and e_j are the surplus variables for the dual.

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Linear Programming

Consider the following primal

$$\text{Max: } z = 60x_1 + 30x_2 + 20x_3$$

s.t.:

$$8x_1 + 6x_2 + x_3 \leq 48$$

$$4x_1 + 2x_2 + 1.5x_3 \leq 20$$

$$2x_1 + 1.5x_2 + 0.5x_3 \leq 8$$

With: $x_1, x_2, x_3 \geq 0$

and its dual

$$\text{Min: } z = 48w_1 + 20w_2 + 8w_3$$

s.t.:

$$8w_1 + 4w_2 + 2w_3 \geq 60$$

$$6w_1 + 2w_2 + 1.5w_3 \geq 30$$

$$w_1 + 1.5w_2 + 0.5w_3 \geq 60$$

With: $w_1, w_2, w_3 \geq 0$



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Linear Programming

The optimal primal solution is

$$(x_1^*, x_2^*, x_3^*, s_1^*, s_2^*, s_3^*)^T = (2, 0, 8, 24, 0, 0)^T$$

The optimal dual solution is

$$(w_1^*, w_2^*, w_3^*, e_1^*, e_2^*, e_3^*)^T = (0, 10, 10, 0, 5, 0)^T$$

We have

$$s_1^* w_1^* = s_2^* w_2^* = s_3^* w_3^* = 0$$

$$e_1^* x_1^* = e_2^* x_2^* = e_3^* x_3^* = 0$$



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Linear Programming

Sensitivity analysis

If certain changes take place in the parameters (a_{ij} , b_i , c_j) of the original model, would the original optimal solution change?

Analyzing how an optimal solution changes with changes in the model is referred to as sensitivity analysis.

Sensitivity analysis does not require completely resolving the linear program with the new data. Often an optimal tableau for one problem can be used to obtain an optimal solution for a problem where data has been changed slightly.



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Linear Programming

Consider the following maximization program with slack variables.

Maximize: $z = C^T X$

Subject to: $AX = B$

With: $X \geq 0$

The initial simplex tableau is given by

	X^T		
X_0	A^*	I_m	B
	$-C^*{}^T$	0	0

($C_0 = 0$)

where

$$A = [A^* | I_m], \quad C^T = [C^*{}^T | 0]$$



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Linear Programming

The final simplex tableau is of the form

	X^T	
X_B	$SA^* \quad S$	SB
	$y^T A^* - C^{*T} \quad y^T$	$y^T B$

where S is the matrix of coefficients of slack variables of the final tableau. $y^T = C_B^T S$ where C_B is the vector whose elements are the objective function coefficients (including zero for slack variables) for the corresponding elements of X_B .



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Linear Programming

We want to see how changes in the original model would change the numbers in the final simplex tableau (assuming that the same sequence of elementary row operations performed by the simplex method were to be duplicated). Therefore, after making a few simple calculations to revise this tableau, we can check easily whether the original optimal basic feasible solution is now non-optimal (or infeasible). If so, this solution would be used as the initial basic solution to restart the simplex method to find the new optimal solution, if desired.



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Linear Programming

Let $A^* \rightarrow A_c^*$, $B \rightarrow B_c$ and $C^* \rightarrow C_c^*$. The revised final simplex tableau becomes

	X^T	
X_B	$SA_c^* \quad S$	SB_c
	$y^T A_c^* - C_c^{*T} \quad y^T$	$y^T B_c$

Note that the revised final tableau is calculated from y and S (which have not changed).

We will just do the following types of sensitivity analysis

1. Changes in B .
2. Changes in the coefficients of a nonbasic variable.
3. Changes in the coefficients of a basic variable.



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Linear Programming

Consider the following program:

$$\text{Max: } z = 3x_1 + 5x_2$$

Subject to:

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

With: $x_1, x_2 \geq 0$

Model 1



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Linear Programming

This program is put into standard form by introducing slack variables x_3 , x_4 and x_5 , respectively in the constraint inequalities.

The initial simplex tableau is

	x_1	x_2	x_3	x_4	x_5	
x_3	1	0	1	0	0	4
x_4	0	2	0	1	0	12
x_5	3	2	0	0	1	18
	-3	-5	0	0	0	0

Tableau X

where

$$A^* = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{bmatrix}, C^{*T} = [3 \ 5]$$



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Linear Programming

The final simplex tableau obtained is

	x_1	x_2	x_3	x_4	x_5	
x_3	0	0	1	1/3	-1/3	2
x_2	0	1	0	1/2	0	6
x_1	1	0	0	-1/3	1/3	2
	0	0	0	3/2	1	36

Where

$$S = \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{bmatrix}, y^T = [0 \ 3/2 \ 1]$$



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Linear Programming

(a) Changes in B

Suppose that the vector B is changed from

$$B = [4 \ 12 \ 18]^T \rightarrow B_c = [4 \ 12+\Delta \ 18]$$

Find the range of values for Δ for the solution to remain feasible*.

To calculate the allowable range to stay feasible, $SB_c \geq 0$ as can be seen from the revised final simplex tableau.

* We assume that the change in this one B_i value is the only change in the model.



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Linear Programming

	X^T		
X_B	SA^*	S	SB_c
	$y^T A^* - C^{*T}$	y^T	$y^T B_c$

** changes in B only

Thus,

$$SB_c = \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{bmatrix} \begin{bmatrix} 4 \\ 12 + \Delta \\ 18 \end{bmatrix} = \begin{bmatrix} \frac{1}{3}(6 + \Delta) \\ \frac{1}{2}(12 + \Delta) \\ \frac{1}{3}(6 - \Delta) \end{bmatrix}$$



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Linear Programming

$$\frac{1}{3}(6 + \Delta) \geq 0 \Rightarrow \Delta \geq -6$$

$$\frac{1}{2}(12 + \Delta) \geq 0 \Rightarrow \Delta \geq -12$$

$$\frac{1}{3}(6 - \Delta) \geq 0 \Rightarrow \Delta \leq 6$$

The solution remains feasible only if

$$-6 \leq \Delta \leq 6$$

Thus, the second component in **B** can only assume values from 6 to 18, i.e.,

$$6 \leq b_2^* \leq 18$$



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Linear Programming

Consider the following program:

$$\text{Max: } z = 3x_1 + 5x_2$$

Subject to:

$$x_1 \leq 4$$

$$2x_2 \leq 24$$

$$3x_1 + 2x_2 \leq 18$$

With: $x_1, x_2 \geq 0$

Model 2



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Linear Programming

The initial Simplex Tableau is similar to Tableau X, except that $B = [4, 24, 18]^T$.

The final simplex tableau is

	x_1	x_2	x_3	x_4	x_5	
x_3	1	0	1	0	0	4
x_4	-1	0	0	1	-1	6
x_2	3/2	1	0	0	1/2	9
	9/2	0	0	0	5/2	45

Tableau Y



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Linear Programming

(b) Changes in the coefficients of a nonbasic variable*

Note that x_1 is a nonbasic variable in the optimal solution shown by the final simplex tableau.

* Because the variables involved is nonbasic (value of zero), changing its coefficients cannot affect the feasibility of the solution. The question now is whether it is still optimal.

Then the only change in A^* is that one or more of the coefficients of x_1 , i.e, c_1^* , a_{11}^* , a_{21}^* , a_{31}^* , have been changed.



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Linear Programming

(i) Suppose the only change is

$$c_1^* = 3 \rightarrow 3 + \Delta$$

Find the allowable range to stay optimal for c_1 (it is assumed that the change in this one c_1 is the only change in the model)

The revised final simplex tableau is

	X^T		
X_B	SA^*	S	SB
	$y^T A^* - C_c^{*T}$	y^T	$y^T B$

**change in C^* only



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Linear Programming

Consider the row vector $y^T A^* - C_c^{*T}$. The first element of the row is

$$y^T A^* - C_{c1}^{*T} = \begin{bmatrix} 0 & 0 & 5/2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} - (3 + \Delta) = 9/2 - \Delta$$

Thus, the revised final simplex tableau is

	x_1	x_2	x_3	x_4	x_5	
x_3	1	0	1	0	0	4
x_4	-3	0	0	1	-1	6
x_2	3/2	1	0	0	1/2	9
	$9/2 - \Delta$	0	0	0	5/2	45

For optimality,

$$9/2 - \Delta \geq 0 \text{ or } \Delta \leq 9/2$$

The allowable range of values is

$$c_1^* \leq 15/2$$

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Linear Programming

(ii) Suppose the changes are

$$A_1^* = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} \rightarrow A_{c1}^* = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$c_1^* = 3 \rightarrow c_{c1}^* = 4$$

Then

$$SA_{c1}^* = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$y^T A_{c1}^* - C_{c1}^* = \begin{bmatrix} 0 & 0 & 5/2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - 4 = 1 > 0$$

$\Rightarrow$ Optimality of the solution.



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Linear Programming

(c) Changes in the coefficients of a basic variable

Consider the final simplex tableau X . x_2 is a basic variable in the optimal solution shown by the final simplex tableau. Then the only changes in the current model 2 are made to the coefficient of this variable.

(i) Suppose that the changes are

$$c_2^* = 5 \rightarrow c_{c2}^* = 3$$

$$A_2^* = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \rightarrow A_{c2}^* = \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix}$$

Then the revised final Simplex Tableau is

	X^T		
X_B	SA_{c}^*	S	SB
	$y^T A_{c}^* - C_{c}^{*T}$	y^T	$y^T B$



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Linear Programming

$$y^T A_{c2}^* - c_{c2}^* = [0 \ 0 \ 5/2] \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix} - 3 = 7$$

$$SA_{c2}^* = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$$

The resulting revised final tableau is

	x_1	x_2	x_3	x_4	x_5	
x_3	1	0	1	0	0	4
x_4	-3	-1	0	1	-1	6
x_2	3/2	2	0	0	1/2	9
	9/2	7	0	0	5/2	45



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Linear Programming

The basic variable x_2 must have a coefficient of 1 in its row of the simplex tableau and a coefficient of 0 in every other row (including the last row), i. e., the proper form from Gaussian elimination.

We must perform elementary row operations to restore the proper form from Gaussian elimination.

	x_1	x_2	x_3	x_4	x_5	
x_3	1	0	1	0	0	4
x_4	-3	-1	0	1	-1	6
x_2	3/4	1	0	0	1/4	9/2
	9/2	7	0	0	5/2	45

$1/2R_3$



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Linear Programming

	x_1	x_2	x_3	x_4	x_5	
x_3	1	0	1	0	0	4
x_4	-9/4	0	0	1	-3/4	21/2
x_2	3/4	1	0	0	1/4	9/2
	-3/4	0	0	0	3/4	27/2

$R_3 + R_2$

$7R_3 + R_4$

The new value of the current basic solution is $x_3 = 4$, $x_4 = 21/2$, $x_2 = 9/2$ ($x_1 = 0$, $x_5 = 0$). Since all these variables are nonnegative, the solution is still feasible. However, because of the negative coefficient of x_1 in the last row, we know that it is no longer optimal.

Applying the simplex method to this tableau gives



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Linear Programming

	x_1	x_2	x_3	x_4	x_5	
x_3	1*	0	1	0	0	4
x_4	-9/4	0	0	1	-3/4	21/2
x_2	3/4	1	0	0	1/4	9/2
	-3/4	0	0	0	3/4	27/2

Ratio

$4/1 = 4$

$(9/2)/(3/4)$



	x_1	x_2	x_3	x_4	x_5	
x_1	1	0	1	0	0	4
x_4	0	0	9/4	1	-3/4	39/2
x_2	0	1	-3/4	0	1/4	3/2
	0	0	3/4	0	3/4	33/2

$9/4R_1 + R_2$

$-3/4R_1 + R_3$

$3/4R_1 + R_4$

The new optimal solution is $x_1^* = 4$, $x_2^* = 3/2$, $x_4^* = 39/2$ ($x_3^* = 0$, $x_5^* = 0$).



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Linear Programming

Consider the following program:

$$\text{Max: } z = 3x_1 + 3x_2$$

Subject to:

$$x_1 \leq 4$$

$$3x_2 \leq 24$$

$$3x_1 + 4x_2 \leq 18$$

With: $x_1, x_2 \geq 0$

Model 3



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Linear Programming

The initial simplex tableau is similar to Tableau X, except that $B = [4 \ 24 \ 18]^T$ and $A_2^* = [0 \ 3 \ 4]^T$, $C^{*T} = [3 \ 3]$.

The final simplex tableau obtained is

	x_1	x_2	x_3	x_4	x_5	
x_1	1	0	1	0	0	4
x_4	0	0	9/4	1	-3/4	39/2
x_2	0	1	-3/4	0	1/4	3/2
	0	0	3/4	0	3/4	33/2

Suppose we want to find the allowable range to stay optimal for c_2 . Note that x_2 is a basic variable for the optimal solution (with $c_2^* = 3$).



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Linear Programming

$$\text{Let } c_2^* = 3 \rightarrow c_{c2}^* = 3 + \Delta$$

The revised final simplex tableau is

	X^T		
X_B	SA^*	S	SB
	$y^T A^* - C_c^*{}^T$	y^T	$y^T B$

** change in C^* only

$$y^T A_2^* - c_{c2}^* = \begin{bmatrix} \frac{3}{4} & 0 & \frac{3}{4} \end{bmatrix} \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix} - (3 + \Delta) = -\Delta$$



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Linear Programming

Thus, we have

	x_1	x_2	x_3	x_4	x_5	
x_1	1	0	1	0	0	4
x_4	0	0	9/4	1	-3/4	39/2
x_2	0	1	-3/4	0	1/4	3/2
	0	$-\Delta$	3/4	0	3/4	33/2

Convert to proper form from Gaussian elimination.



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Linear Programming

	x_1	x_2	x_3	x_4	x_5	
x_1	1	0	1	0	0	4
x_4	0	0	9/4	1	-3/4	39/2
x_2	0	1	-3/4	0	1/4	3/2
	0	0	$-3/4\Delta + 3/4$	0	$1/4\Delta + 3/4$	$3/2\Delta + 33/2$

$\Delta R_3 + R_4$

To keep the coefficients of the nonbasic variables (x_3 and x_5) nonnegative,

$$-3/4\Delta + 3/4 \geq 0 \Rightarrow \Delta \leq 1$$

$$1/4\Delta + 3/4 \geq 0 \Rightarrow \Delta \geq -3$$

This means the allowable range of values is

$$-3 \leq \Delta \leq 1$$

Thus, the allowable range to stay optimal for c_2^* is

$$0 \leq c_2^* \leq 4$$



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Linear Programming

Duality and sensitivity analysis

This section shows how knowledge of duality can be applied to sensitivity analysis.

Consider this result. *Assuming that a set of basic variables (for the primal) is feasible, then it is optimal if and only if the associated dual solution is dual feasible.*

This result can be used for an alternative way of doing the following types of sensitivity analysis:

1. Changing the objective function coefficient of a nonbasic variable.
2. Changing the column for a nonbasic variable.

In each case, we determine whether a change in the *original* LP maintains dual feasibility. If dual feasibility is not maintained, the current *primal* solution is no longer optimal.



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Linear Programming

Consider the following program:

$$\text{Max: } z = 60x_1 + 30x_2 + 20x_3$$

s.t.:

$$8x_1 + 6x_2 + x_3 \leq 48$$

$$4x_1 + 2x_2 + 1.5x_3 \leq 20$$

$$2x_1 + 1.5x_2 + 0.5x_3 \leq 8$$

With: $x_1, x_2, x_3 \geq 0$

and its dual

$$\text{Min: } z = 48w_1 + 20w_2 + 8w_3$$

s.t.:

$$8w_1 + 4w_2 + 2w_3 \geq 60$$

$$6w_1 + 2w_2 + 1.5w_3 \geq 30$$

$$w_1 + 1.5w_2 + 0.5w_3 \geq 20$$

With: $w_1, w_2, w_3 \geq 0$



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Linear Programming

The primal solution is $(x_1^*, x_2^*, x_3^*)^T = (2, 0, 8)^T$.

The primal dual solution is $(w_1^*, w_2^*, w_3^*)^T = (0, 10, 10)^T$.

(a) Changing the objective function coefficient of a nonbasic variable.

Let c_2 be the coefficient of x_2 in the objective function. For what values of c_2 will the current solution x^* remain optimal?

If $w_1^* = 0, w_2^* = 10, w_3^* = 10$ remains dual feasible, the current solution is unchanged.

Note that if the objective function coefficient for x_2 is changed, the first and third dual constraints remain unchanged, the first and third dual constraints remain unchanged, but the second dual constraint is changed to

$$6w_1 + 2w_2 + 1.5w_3 \geq c_2$$

If $w_1^* = 0, w_2^* = 10, w_3^* = 10$ satisfies this inequality, dual



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Linear Programming

feasibility (and therefore primal optimality) is maintained. Thus, the current solution remains optimal if c_2 satisfies

$$6(0) + 2(10) + 1.5(10) \geq c_2$$

or

$$c_2 \leq 35$$

This shows that for $c_2 \leq 35$, the current solution x^* remains optimal. Conversely, if $c_2 > 35$, the current basis is no longer optimal.



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Linear Programming

(b) Changing the column for a nonbasic variable

Suppose

$$a_{12} = 6 \rightarrow 5, a_{32} = 2 \rightarrow 1.5, c_2 = 30 \rightarrow 43$$

Changing the column for the nonbasic variable x_2 leaves the first and third dual constraints unchanged but changes the

Second dual constraint:

$$6w_1 + 2w_2 + 1.5w_3 \geq 30 \rightarrow 5w_1 + 2w_2 + 2w_3 \geq 43$$

Since $w_1^* = 0, w_2^* = 10, w_3^* = 10$,

$$5(0) + 2(10) + 2(10) = 40 \text{ not } \geq 43$$

Thus, the dual feasibility is not maintained and the current solution is no longer optimal.



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EE6204 Systems Analysis

Lecture 3: Special Types of LP

presented by

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Special Types of LP Problems

- **Special Types of LP Problems**

These LP problems require a very large number of constraints and variables. An application of the simplex method may require an exorbitant computational effort. However, the special structures of problems can be exploited to develop more efficient algorithms.



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Special Types of LP Problems

• Transportation model

The transportation problem is concerned with shipping any commodity from any group of supply centers, called sources, to any group of receiving centers, called destinations, in such a way as to minimize the total shipping cost.

In the transportation problem units of a single product are to be shipped from m sources to n destinations where:

s_i = number of units available at source i

d_j = number of units required at destination j

c_{ij} = cost of shipping one unit source i to destination j

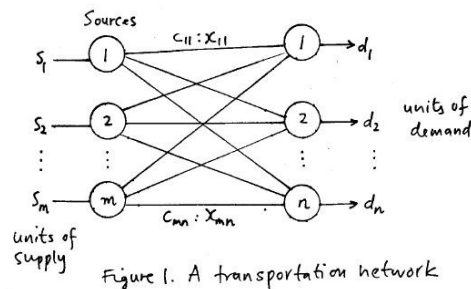


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Special Types of LP Problems

This problem is to determine a shipping program that meets the required demands and minimizes the total shipping cost. The solution must have integer value.

Figure 1 shows the transportation model as a network with m sources and n destinations.



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Special Types of LP Problems

Assuming the total supply is equal to the total demand. Let z be the total shipping cost and x_{ij} be the number of units to be shipped from source i to destination j ; then the linear program representing the transportation problem is

$$\text{minimize } z = \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

Subject to

$$\sum_{j=1}^n x_{ij} = s_i \quad (\text{supply constraints})$$

$$(1) \quad \sum_{i=1}^m x_{ij} = d_j \quad (\text{demand constraints})$$

$$x_{ij} \geq 0$$

Equation (1) is a balanced transportation problem because total supply equals total demands.



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Special Types of LP Problems

Consider the shipment of steel from two warehouses W_1 and W_2 to two markets M_1 and M_2 . The cost of shipping from warehouse W_i to M_j is given in the i th row and j th column of the following table.

	M_1	M_2	Supply
W_1	5	8	15
W_2	4	10	5
Demand	12	8	

For example, the cost of shipping from W_1 to M_2 is $c_{12} = \$8 / \text{ton}$

Note that the sum of the supplies equals the sum of the demands.

$$s_1 + s_2 = 20 = d_1 + d_2$$



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Special Types of LP Problems

Let x_{ij} be the amount of in tons to be shipped from warehouse W_i to market M_j . The problem is to ship the steel in the least expensive (minimum cost) way and in so doing completely exhaust the supplies at the warehouses and exactly satisfies the demands at the markets.

The cost in \$ in shipping x_{11} tons from W_1 to M_1 is 5 times x_{11} , i.e., $5x_{11}$. We can thus express the total shipping cost z in the following way:

$$\min z = 5x_{11} + 8x_{12} + 4x_{21} + 10x_{22}$$



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Special Types of LP Problems

The requirement pertaining to the warehouses and markets are equation constraints:

$$\begin{array}{lcl} x_{11} + x_{12} = 15 & \left. \begin{array}{l} \\ \\ \end{array} \right\} & \text{Supply constraints} \\ x_{21} + x_{22} = 5 & & \\ x_{11} + x_{21} = 12 & \left. \begin{array}{l} \\ \\ \end{array} \right\} & \text{Demand constraints} \\ x_{12} + x_{22} = 8 & & \end{array}$$

and

$$x_{11}, x_{12}, x_{21}, x_{22} \geq 0$$

The problem takes the following form:

$$\min: z = C^T X$$

$$\text{subject to: } AX = B$$

$$\text{with: } X \geq 0$$

where A is a $(m+n) \times (mn)$ matrix and has a special structure.



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Special Types of LP Problems

For the problem we have

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{12} \\ x_{21} \\ x_{22} \end{bmatrix} = \begin{bmatrix} 15 \\ 5 \\ 12 \\ 8 \end{bmatrix}$$

It is the A matrix that gives the transportation problem its special structure.

For many applications, the supply and demand quantities in the model (the s_i and d_j) have integer values, and the implementation will require that the shipping quantities (the x_{ij}) also have integer values. Fortunately, because of the special structure of the transportation model, it has the following property.



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Special Types of LP Problems

- **Integer solutions property**

For transportation problems where every s_i and d_j have integer value, all the basic variables (allocations) in every basic feasible solution (including an optimal one) also have integer values.

A transportation problem must possess feasible solutions. The following property indicates when this will occur.



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Special Types of LP Problems

- Feasible solutions property**

A transportation problem has feasible solutions if and only if the total supply equals the total demand, i.e.,

$$\sum_{i=1}^m s_i = \sum_{j=1}^n d_j \quad (1)$$

Eqn.(1) is guaranteed by creating either a dummy destination with a demand equal to the surplus if the total demand is less than the total supply, or a dummy source with a supply equal to the shortage if the total demand exceeds total supply. The shipping cost from any source to a dummy destination is taken to be zero. Similarly, the shipping cost from a dummy source to any destination is taken to be zero.



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Special Types of LP Problems

- Transportation tableau**

The relevant data (supply, demand, and the shipping costs) for any transportation problem can be summarized in a transportation tableau (Tableau 1). The tableau is a compact method for representing the transportation model.

		Destinations				
		1	2	...	n	Supply s_i
Sources	1	c_{11}	c_{12}	...	c_{1n}	s_1
	2	c_{21}	c_{22}	...	c_{2n}	s_2
	...	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	m	c_{m1}	c_{m2}	...	c_{mn}	s_m
	Demand d_j	d_1	d_2	...	d_n	

Tableau 1



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Special Types of LP Problems

A car rental company is faced with an allocation problem resulting from rental agreements that allow cars to be returned to locations other than those at which they were originally rented. At the present time, there are two locations (sources) with 15 and 13 surplus cars, respectively, and 4 locations (destinations) requiring 9, 6, 7, and 9 cars, respectively. Unit transportation costs (in dollars) between the locations are as follows:

		Destination			
		1	2	3	4
Source	1	45	17	21	30
	2	14	18	19	31

Set up the initial transportation tableau for the minimum-cost schedule



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Special Types of LP Problems

Since the total demand ($9 + 6 + 7 + 9 = 31$) exceeds the total supply ($15 + 13 = 28$), a dummy source is created having a supply equal to the 3-unit shortage. In reality, shipments from this fictitious source are never made, so the associated shipping costs are taken as zero. Positive allocations from this source to a destination represent cars that cannot be delivered due to a shortage of supply; they are shortages a destination will experience under an optimal shipping schedule.

The transportation tableau is shown below.



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Special Types of LP Problems

		Destinations				Supply	u_i
		1	2	3	4		
Sources	1	45	17	21	30	15	
	2	14	18	19	31	13	
	(b) 3	0	0	0	0	3	
	Demand	9	6	7	9		
v_j							

The x_{ij} , u_i and v_j are not entered, since they are not known at the moment.



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Special Types of LP Problems

Eqn. (1) can be solved by the simplex method. However, even for modest values of m and n , the resulting linear program can be very large. For example, when $m = 100$ and $n = 100$, there are $m+n = 200$ constraints and $mn = 10,000$ variables in the model.

The transportation model has a special structure which allows the development of a solution procedure that is computationally more efficient.



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Special Types of LP Problems

• The transportation algorithm

The steps of the algorithm are as follows:

1. Find an initial basic feasible solution X .
2. Test the solution for optimality.
3. Improve the solution when it is not optimal.
4. Repeat steps (2) and (3) until the optimal solution is obtained.

We shall see how each of these steps can be carried out directly on the transportation tableau.

For transportation problems with m sources and n destinations, the number of functional constraints is $m+n$. However, the number of basic variables is $m+n-1$. The procedure for constructing an initial basic feasible solution selects the $m+n-1$ basic variables one at a time.



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Special Types of LP Problems

• Finding an initial basic solution

Vogel's approximation method

Step 1

Calculate a penalty for each row (column) by subtracting the smallest cost element from the next smallest cost element in the same row (column). If two unit costs tie for being the smallest remaining in a row or column, then the difference is 0.

Step 2

Identify the row or column with the largest penalty, breaking ties arbitrarily. Allocate as much as possible to the variable with the least cost in the selected row or column. Adjust the supply and demand and eliminate the satisfied row or column. If a row and column are satisfied simultaneously, only one of them is eliminated and the remaining row (column) is assigned a zero supply (demand).

Step 3

If exactly one row or column remains, stop.



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Special Types of LP Problems

Consider the following transportation tableau.

		Destinations				Supply	u_i
		1	2	3	4		
Sources	1	10	5	1	5	10	
	2	2	5	7	6	15	
	3	12	2	6	13	25	
Demand		8	20	12	10		
v_j							



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Special Types of LP Problems

Use Vogel's approximation method to obtain an initial basic feasible solution.

$$\text{no. of basic variables} = m + n - 1 = 3 + 4 - 1 = 6$$

Demand = Supply $\rightarrow$ no dummy needed.

		1	2	3	4	Supply
	1	10	5	1	5	10
	2	2*	5	7	6	15
	3	12	2	6	13	25
Demand		8	20	12	10	

col. diff. 8+ 3 5 1
 $\uparrow$
 col. with largest penalty

row diff.

- 4 * Variable with least cost in col.
- 3
- 4 select $x_{21}=8$
 Eliminate col 1
 (demand met)



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Special Types of LP Problems

	1	2	3	4	Supply	row diff.
1	5		1*	5	10	4
2	5		7	6	7	1
3	2		6	13	25	4
Demand	20	12	10			
col. diff.	3	5+	1			

Select $x_{13} = 10$
Eliminate row 1
(Supply 1 exhausted)



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Special Types of LP Problems

	1	2	3	4	Supply	row diff.
2	5		7	6*	7	1
3	2		6	13	25	4
Demand	20	2	10			
col. diff.	3	1	7+			

Select $x_{24} = 7$
Eliminate row 2
(Supply 2 exhausted)

	1	2	3	4	Supply
3	2		6	13	25
Demand	20	2	3		

Select
 $x_{32} = 20$
 $x_{33} = 2$
 $x_{34} = 3$



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Special Types of LP Problems

The allocation is shown in the following tableau.

	1	2	3	4	Supply	u_i
1	10	5	1	5	10	
2	2	8	7	6	15	
3	12	2	6	13	25	
Demand	8	20	12	10		
v_j						

Tableau A

The circled numbers are the values of the basic variables.

Thus, $x_{13}=10$, $x_{21}=8$, $x_{24}=7$, $x_{32}=20$, $x_{33}=2$, $x_{34}=3$

with other variables nonbasic and equal to 0.

The value of z for this solution is

$$Z = c_{13}x_{13} + c_{21}x_{21} + c_{24}x_{24} + c_{32}x_{32} + c_{33}x_{33} + c_{34}x_{34}$$

$$= 1 \cdot 10 + 2 \cdot 8 + 6 \cdot 7 + 2 \cdot 20 + 6 \cdot 2 + 13 \cdot 3 = 159 \quad (\text{Is this optimal?})$$



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Special Types of LP Problems

Test for optimality

Assign one (any one) of the u_i or v_j in Tableau 1 the value 0 and calculate the remaining u_i and v_j so that for each basic variable $u_i + v_j = c_{ij}$.

Then, for each nonbasic variable, calculate the quantity $c_{ij} - u_i - v_j$.

If all $c_{ij} - u_i - v_j \geq 0$, the current solution is optimal; otherwise, the current solution is not optimal.

If the current basic feasible solution is not optimal, the transportation method must next go to an iteration to find a better feasible solution.



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Special Types of LP Problems

There are 2 steps involved:

1. Calculate u_i and v_j with respect to the basic variables.
 2. Compute $c_{ij} - u_i - v_j$ for each nonbasic variables cell.
1. Select the u_i that has the largest number of allocations in its row (break any tie arbitrarily) and assign to it the value 0. This choice will simplify computations.

Choose $u_3 = 0$

$$x_{32} - (3,2) \text{ cell: } u_3 + v_2 = c_{32}, 0 + v_2 = 2 \rightarrow v_2 = 2$$

$$x_{33} - (3,3) \text{ cell: } u_3 + v_3 = c_{33}, 0 + v_3 = 6 \rightarrow v_3 = 6$$

$$x_{34} - (3,4) \text{ cell: } u_3 + v_4 = c_{34}, 0 + v_4 = 13 \rightarrow v_4 = 13$$

$$x_{24} - (2,4) \text{ cell: } u_2 + v_4 = c_{24}, u_2 + 13 = 6 \rightarrow u_2 = -7$$

$$x_{21} - (2,1) \text{ cell: } u_2 + v_1 = c_{21}, -7 + v_1 = 2 \rightarrow v_1 = 9$$

$$x_{13} - (1,3) \text{ cell: } u_1 + v_3 = c_{13}, u_1 + 6 = 1 \rightarrow u_1 = -5$$



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Special Types of LP Problems

2. Now calculate the quantities $c_{ij} - u_i - v_j$ for each nonbasic variables cell.

$$x_{11} - (1,1) \text{ cell: } c_{11} - u_1 - v_1 = 10 - (-5) - 9 = 6$$

$$x_{12} - (1,2) \text{ cell: } c_{12} - u_1 - v_2 = 5 - (-5) - 2 = 8$$

$$x_{14} - (1,4) \text{ cell: } c_{14} - u_1 - v_4 = 5 - (-5) - 13 = -3$$

$$x_{22} - (2,2) \text{ cell: } c_{22} - u_2 - v_2 = 5 - (-7) - 2 = 10$$

$$x_{23} - (2,3) \text{ cell: } c_{23} - u_2 - v_3 = 7 - (-7) - 6 = 8$$

$$x_{31} - (3,1) \text{ cell: } c_{31} - u_3 - v_1 = 12 - 0 - 9 = 3$$



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Special Types of LP Problems

The results are recorded in the following tableau.

	1	2	3	4	Supply	u_i
1	10 6	5 8	1 10	5 -3	10	-5
2	2 8	5 10	7 8	6 7	15	-7
3	12 3	2 20	6 2	13 3	25	0
Demand	8	20	12	10		
v_j	9	2	6	13		

Since one of these $c_{ij} - u_i - v_j < 0$, the current solution is NOT optimal.



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Special Types of LP Problems

Improving the solution

A loop is a sequence of cells in Tableau 1 such that: (1) each pair of consecutive cells lie in either the same row or the same column; (2) no 3 consecutive cells lie in the same row or column; (3) the first and last cells of the sequence lie in the same row or column; (4) no cells appear more than once in the sequence.

	1	2	3	4	5	6
1		•	→	•		
2		↑		•	→	•
3			↓			•
4		•	←	•		

Sequence $\{(1,2), (1,4), (3,4), (3,6), (4,6), (4,2)\}$ is a loop.

	1	2	3	4	5	6
1				•	→	•
2	•	•	•	•	•	↓
3	•	•	←	•	←	•
4	•	•	•	•		

Note that a row or column can have more than 2 cells in the loop (2nd row), but no more than 2 can be consecutive.



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Special Types of LP Problems

Consider the nonbasic variable corresponding to the most negative of $c_{ij} - u_i - v_j$ calculated in the test for optimality; it is made the incoming variable. Construct a loop consisting exclusively of this incoming variable (cell) and current basic variables (cells). Then allocate to the incoming cell as many units as possible such that, after appropriate adjustments have been made to the other cells in the loop, the supply and demand constraints are not violated, all allocations remaining nonnegative, and one of the old basic variables has been reduced to 0 (whereupon it ceases to be basic).



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Special Types of LP Problems

Degeneracy

A non degenerate basic feasible solution will be characterized by positive values for $m+n-1$ basic variables. If the process of improving the current basic solution results in 2 or more current basic variables being reduced to 0 simultaneously, only 1 is allowed to become nonbasic. The other variables remain basic, but with a 0 allocation, thereby rendering the basic solution degenerate.



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Special Types of LP Problems

Improving the solution

Take the most negative $c_{ij} - u_i - v_j$ as the incoming variable, i.e., cell(1,4). Identify a loop containing this cell with only basic variable cells. The basic variables to be decreased to 0 then becomes the leaving basic variable.

The loop chosen is shown. x_{34} removed from our set of basic variables.

				Supply
	5	1	5	10
	5	7	6	25
	2	6	13	25
Demand	20	12	10	

Handwritten annotations: A loop is identified starting from cell (1,4) with a '+' sign, moving left to cell (1,2), then down to cell (2,2), then down to cell (3,2), then right to cell (3,3), then up to cell (2,3), and finally back to cell (1,4). The value in cell (1,2) is circled as 10, and the value in cell (3,3) is circled as 3. The value in cell (1,4) is circled as 5.

Special Types of LP Problems

The new basic solution is shown in Tableau B.

	1	2	3	4	Supply	u_i
1	10	5	1	5	10	
2	2	5	7	6	15	
3	12	2	6	13	25	
Demand	8	20	12	10		
v_j						

Handwritten annotations: In row 1, the value 10 is circled. In row 2, the value 2 is circled. In row 3, the value 2 is circled. In row 1, the value 1 is circled. In row 2, the value 7 is circled. In row 3, the value 6 is circled. In row 1, the value 5 is circled. In row 2, the value 6 is circled. In row 3, the value 13 is circled.

We now check whether the solution is optimal. The result is shown in Tableau C.

Special Types of LP Problems

	1	2	3	4	Supply	u_i
1	10 9	5 8	11 7	5 3	10	-5
2	2 8	5 7	7 5	6 7	15	-4
3	12 6	2 20	8 5	13 3	25	0
Demand	8	20	12	10		
v_j	6	2	6	10		

Tableau C.

It is seen that each $c_{ij} - u_i - v_j$ is nonnegative. Hence the new solution is optimal. That is,

$$x_{13}^* = 7, x_{14}^* = 3, x_{21}^* = 8, x_{24}^* = 7, x_{32}^* = 20, x_{33}^* = 5,$$

with all other variables nonbasic and equal to zero.

$$\text{Furthermore, } Z' = c_{13}x_{13} + c_{14}x_{14} + c_{21}x_{21} + c_{24}x_{24} + c_{32}x_{32} + c_{33}x_{33} \\ = 1 \cdot 7 + 5 \cdot 3 + 2 \cdot 8 + 6 \cdot 7 + 2 \cdot 20 + 6 \cdot 5 = 150$$



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Special Types of LP Problems

• The assignment problem

The assignment problem is a special type of LP problem where assignees are being assigned to perform tasks. For example, the assignees might be employees who need to be given work assignments. However, assignees need not be people. They could also be machines, or vehicles, or plants, or even time-slots to be assigned tasks.

The assignment problem is usually stated as follows:

Given n people to be assigned to n jobs, with c_{ij} being the cost of assigning person i to job j , find an assignment of minimum total cost.



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Special Types of LP Problems

To fit the definition of an assignment problem, the applications need to be formulated in a way that satisfy the following assumptions:

1. The no. of assignees and the no. of tasks are the same. (Say, n .)
2. Each assignee is to be assigned to exactly one task.
3. Each task is to be assigned to exactly one assignee.
4. There is a cost c_{ij} associated with assignee i ($i=1,2,\dots,n$) performing task j ($j=1,2,\dots,n$).
5. The objective is to determine how all n assignments should be made in order to minimize the cost.

Dummy assignees or *dummy tasks* can be used to satisfy the first 3 assumptions



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Special Types of LP Problems

The Assignment problem model

The assignment problem uses the following decision variables:

$$x_{ij} = \begin{cases} 1 & \text{if assignee } i \text{ performs task } j \\ 0 & \text{if not} \end{cases}$$

By letting z denote the total cost, the assignment problem model is

$$\text{minimize } z = \sum_{i=1}^n \sum_{j=1}^n c_{ij} x_{ij}$$

Subject to

$$\sum_{j=1}^n x_{ij} = 1, \quad i = 1, \dots, n$$

$$\sum_{i=1}^n x_{ij} = 1, \quad j = 1, \dots, n$$

$$x_{ij} \geq 0 \text{ for all } i \text{ and } j \text{ (} x_{ij} \text{ binary, for all } i \text{ and } j \text{)}$$

******In general, an assignment problem is a balanced transportation problem in which all supplied and demands equal to 1. The sources now are assignees and the destinations now are tasks



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Special Types of LP Problems

Machineco has 4 machines and 4 jobs to be completed. Each machine must be assigned to complete one job. The time required to set up each machine for completing each job is shown in the Table.

	Time (hours)			
	Job 1	Job 2	Job 3	Job 4
Machine 1	14	5	8	7
Machine 2	2	12	6	5
Machine 3	7	8	3	9
Machine 4	2	4	6	10

Setup times for Machineco

Machineco wants to minimize the total setup time needed to complete the 4 jobs.



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Special Types of LP Problems

Machineco must determine which machine should be assigned to each job. We define (for $i, j = 1, 2, 3, 4$)

$x_{ij} = 1$ if machine i is assigned to meet the demands of job j

$x_{ij} = 0$ if machine i is not assigned to meet the demands of job j .

Then Machineco's problem may be formulated as

$$\begin{aligned} \text{Min } z = & 14x_{11} + 5x_{12} + 8x_{13} + 7x_{14} + \\ & 2x_{21} + 12x_{22} + 6x_{23} + 5x_{24} + \\ & 7x_{31} + 8x_{32} + 3x_{33} + 9x_{34} + \\ & 2x_{41} + 4x_{42} + 6x_{43} + 10x_{44} \end{aligned}$$

s.t.:

$$\begin{aligned} x_{11} + x_{12} + x_{13} + x_{14} &= 1 \\ x_{21} + x_{22} + x_{23} + x_{24} &= 1 \\ x_{31} + x_{32} + x_{33} + x_{34} &= 1 \\ x_{41} + x_{42} + x_{43} + x_{44} &= 1 \end{aligned}$$

machine constraints



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Special Types of LP Problems

$$\left. \begin{aligned} x_{11} + x_{12} + x_{13} + x_{14} &= 1 \\ x_{12} + x_{22} + x_{32} + x_{42} &= 1 \\ x_{13} + x_{23} + x_{33} + x_{43} &= 1 \\ x_{14} + x_{24} + x_{34} + x_{44} &= 1 \end{aligned} \right\} \text{job constraints}$$

with: $x_{ij} = 0$ or $x_{ij} = 1$

If $x_{ij} = 1$, the objective function will pick up the time required to set up machine i for job j ; if $x_{ij} = 0$, the objective function will not pick up the time required to set up machine i for job j .

For any assignment problem, practitioners do not bother writing out the full mathematical model. It is simpler to formulate the problem by filling out a cost table (Tableau 2), including identifying the assignees and tasks, since this table contains all the essential data in a far more compact form



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Special Types of LP Problems

	1	2	...	n
1	c_{11}	c_{12}	...	c_{1n}
2	c_{21}	c_{22}	...	c_{2n}
...	...	...	...	...
n	c_{n1}	c_{n2}	...	c_{nn}

Tableau 2

Hungarian method

The transportation algorithm can be used to solve assignment problems. But a solution method more efficient than the transportation algorithm is the *Hungarian* method, which operates directly on the cost matrix. It converts the original cost table into a series of *equivalent* cost tables until it reaches one where an optimal solution is obvious. This final equivalent cost table is the one consisting of only *positive or zero* elements where all the assignments can be made to the zero element positions, This set of assignments with a zero total cost is optimal.

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Special Types of LP Problems

There are four steps:

Step 1.

In each row of Tableau 2, locate the smallest element and subtract it from every element in that row. Repeat this procedure for each column (the column minimum is determined after the row subtractions). The revised cost matrix will have at least one zero in every row and column.

Step 2.

Determine whether there exists a feasible assignment involving only zero costs in the revised cost matrix. In other words, find if the revised matrix has n zero entries no two of which are in the same row or column. If such assignment exists, it is optimal. If not, go to step 3.

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Special Types of LP Problems

Step 3.

Cover all zeros in the revised matrix with as few horizontal and vertical lines as possible. Each horizontal (vertical) line must pass through an entire row (column). The total number of lines in this minimal covering will be less than n . Locate the smallest number from every element not covered by a line. Subtract this number from every element not covered by a line and add it to every element that lies at the intersection of two lines.

Step 4.

Return to step 2.



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Special Types of LP Problems

Example

4 employees are available to perform 4 jobs. The costs for filling the jobs are given in the following table. Determine the assignment of employees to jobs that minimize the total costs for filling the jobs.

	J_1	J_2	J_3	J_4
P_1	-	30	23	25
P_2	34	-	16	24
P_3	22	19	21	-
P_4	21	23	14	20

* Dashes indicate person cannot do that particular job.



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Special Types of LP Problems

If it is not possible to assign the i th applicant to the j th job, we set cost c_{ij} equal to M , an exceedingly large positive cost.

** For an assignment problem, the cost c_{ij} is the total cost associated with assignee i performing task j .

We now have the following cost matrix.

	1	2	3	4
1	M	30	23	25
2	34	M	16	24
3	22	19	21	M
4	21	23	14	20

Table A

$n = m = 4$ assignments to be made.



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Special Types of LP Problems

We now employ Hungarian method.

	1	2	3	4
1	M	7	0	2
2	18	M	0	8
3	3	0	2	M
4	7	9	0	6

- Subtract 23, 16, 19 and 14 from rows 1-4

$M - 19 \rightarrow M$



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Special Types of LP Problems

	1	2	3	4
1	M	7	0	0
2	15	M	0	6
3	0	0	2	M
4	4	9	0	4

- Subtract 3, 0, 0 and 2 from columns 1-4.
- Only 3 assignments to 0 element positions possible.



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Special Types of LP Problems

	1	2	3	4
1	M	7	0	0
2	15	M	0	6
3	0	0	2	M
4	4	9	0	4

- Cover all 0's
- The minimum element not crossed out is 4



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Special Types of LP Problems

	1	2	3	4
1	M	7	4	0*
2	11	M	0*	2
3	0	0*	6	M
4	0*	5	0	0

- Subtract every element not covered by a line by 4
- Add 4 to the elements at the intersections of 2 lines.
- Now 4 assignments to 0 element positions possible.



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Special Types of LP Problems

The set of zeros is indicated by the *. The resulting total cost is seen
From the cost matrix Table A to be

$$\begin{aligned} Z^* &= c_{14} + c_{23} + c_{32} + c_{41} \\ &= 25 + 16 + 19 + 21 = 81. \end{aligned}$$

Optimal assignment is:

Assign P_1 to J_4 , P_2 to J_3 , P_3 to J_2 and P_4 to J_1 .



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Special Types of LP Problems

Example

Suppose that there are 6 people applying for 5 jobs, and it is desired to fill each job with exactly one person. The costs for filling the jobs with 6 people are given in the following table:

	J_1	J_2	J_3	J_4	J_5
P_1	27	23	22	24	27
P_2	28	27	21	26	24
P_3	28	26	24	25	28
P_4	27	25	21	24	24
P_5	25	20	23	26	26
P_6	26	21	21	24	27

Determine the optimal plan whereby the total cost of assigning the people is minimized.



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Special Types of LP Problems

Here, the number of people exceeds the number of jobs. We introduce a dummy job J_6 . As usual, costs associated with dummy job J_6 are taken to be 0. Person i assigned to J_6 means, in effect, that person i will not be assigned. We now have the following table.

	J_1	J_2	J_3	J_4	J_5	J_6
P_1	27	23	22	24	27	0
P_2	28	27	21	26	24	0
P_3	28	26	24	25	28	0
P_4	27	25	21	24	24	0
P_5	25	20	23	26	26	0
P_6	26	21	21	24	27	0

Table A



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Special Types of LP Problems

There are $m = n = 6$ assignments to be made.

Applying the Hungarian method results in the following revised cost matrix with 6 0's no two of which are in the same row or column.

	1	2	3	4	5	6
1	1	2	1	0*	3	0
2	2	6	0*	2	0	0
3	2	5	3	1	4	0*
4	1	4	0	0	0*	0
5	0*	0	3	3	3	1
6	0	0*	0	0	3	0



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Special Types of LP Problems

There are many ways of making a complete assignment. One of them is shown by the 6*'s. The resulting total cost is seen from Table A to be

$$\begin{aligned} Z^* &= c_{14} + c_{23} + c_{36} + c_{45} + c_{51} + c_{62} \\ &= 24 + 21 + 0 + 24 + 25 + 21 \\ &= 115 \end{aligned}$$

The other complete assignment is colored green in the table.

$$\begin{aligned} Z^* &= c_{14} + c_{25} + c_{36} + c_{43} + c_{52} + c_{61} \\ &= 24 + 24 + 0 + 21 + 20 + 26 \\ &= 115 \end{aligned}$$

Note that the assignment of P_3 to J_6 means that P_3 will not be assigned.



EE6204 Systems Analysis

Lecture 4: Optimization Techniques

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Nonlinear Programming

- **Nonlinear Programming**

In the previous lectures, we studied linear programming(LP) problems. LP problems are concerned with finding the maximum or minimum of a *linear* function

$$Z = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

subject to *linear* constraints on the variables

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$\dots\dots\dots a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

where all $x_j \geq 0$ and c_j , b_i and a_{ij} are known constants.

In many real life optimization problems, the objective function may not be a linear function or some of the constraints may not be linear constraints. Such optimization problems are called nonlinear programming (NLP) problems.

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Nonlinear Programming

The NLP problem is to find $X = (x_1, x_2, \dots, x_n)^T$ to
minimize $f(X)$

subject to

$$h_i'(X) = \alpha_i, \text{ for } i = 1, 2, \dots, m$$

$$g_j'(X) \leq \beta_j, \text{ for } j = 1, 2, \dots, p$$

where $f(X)$, $h_i'(X)$ and $g_j'(X)$ are given (NL) functions of the n decision variables. $f(X)$ is the NLP's objective function and $h_i'(X) = \alpha_i$, $g_j'(X) \leq \beta_j$ are the NLP's *constraints*.

In vector notation, the above problem can be represented in the following standard form:

Minimize $f(X)$

Subject to:

$$h(X) = h'(X) - \alpha = 0$$

$$g(X) = g'(X) - \beta \leq 0$$



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Nonlinear Programming

Example

Consider the following program:

$$\text{Min } (x_1 - 1)^2 + x_2 - 2$$

Subject to:

$$x_2 - x_1 = 1$$

$$x_1 + x_2 \leq 2$$

The problem is already in the standard form with

$$f(x_1, x_2) = (x_1 - 1)^2 + x_2 - 2$$

$$h(x_1, x_2) = x_2 - x_1 - 1$$

$$g(x_1, x_2) = x_2 + x_1 - 2$$



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Nonlinear Programming

Definition

Any point satisfying the constraints is called a *feasible point*.

The set of all feasible points $\{X: h(X) = 0, g(X) \leq 0\}$ is called the feasible set.

An NLP with no constraints is known as an unconstrained NLP.

We can minimize $f(X)$ by solving

$$\text{Maximize } -f(X)$$

and then multiply the optimal objective value by -1.

Note that there does not exist an algorithm that solves every specific problem fitting the format given above.

In this course, we will look at some of the basic ideas for solving certain important types of NLP problems.



5

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Nonlinear Programming

Example

It costs a company \$ c per unit to manufacture a product. If the company charges \$ p per unit for the product, customers demand $D(p)$ units. In order to maximize profits, what price should the firm charge?

The firm's decision variable is p . Since the firm's profit is $(p-c)D(p)$, the firm wishes to solve the following unconstrained maximization problem: $\max (p-c)D(p)$.



6

6

Nonlinear Programming

Example

If K units of capital and L units of labor are used, a company can produce KL units of a manufactured good. Capital can be purchased at \$4/unit and labor can be purchased at \$1/unit. A total of \$8 is available to purchase capital and labor. How can the firm maximize the quantity of the good that can be manufactured?

Let K = units of capital purchased and L = units of labour purchased. Then K and L must satisfy $4K + L \leq 8$, $K \geq 0$, and $L \geq 0$. Thus, the firm wished to solve the following constrained maximization problem:

$$\text{Max } f = KL$$

Subject to:

$$4K + L \leq 8$$

$$K, L \geq 0$$



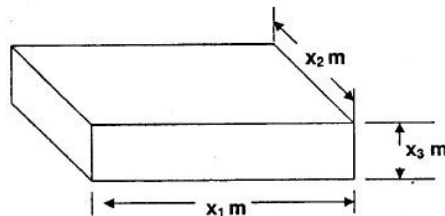
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Nonlinear Programming

Example

400 m³ of granular product are to be transported across a river by ferry. Any amount of the product can be transported in one trip, and regardless of how much is transported at one time, the cost is \$0.10 per trip.

The granular product is to be carried by a container (see figure) and transported across the river within 300 trips.



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Nonlinear Programming

Cost per m^2 for the sides, bottom, and ends of the container are as follows.

1. Ends: $\$20.00/m^2$
2. Bottom: $\$10.00/m^2$
3. Sides: $\$10.00/m^2$

We would like to know what sizes of x_1 , x_2 and x_3 will minimize the sum of the container cost and the transport cost.

Our costs are as follows:

Container costs:

$$\text{Ends: } (2)(\text{width})(\text{depth})(\$20) = 40x_2x_3$$

$$\text{Bottom: } (\text{length})(\text{width})(\$10) = 10x_1x_2$$

$$\text{Sides: } (2)(\text{length})(\text{depth})(\$10) = 20x_1x_3$$



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Nonlinear Programming

Transport cost:

(No. of trips)(cost per trip)

$$= 400/(x_1x_2x_3)(\$0.10)$$

$$= 40/(x_1x_2x_3)$$

The total cost f is given by

$$f(x_1, x_2, x_3) = 40/(x_1x_2x_3) + 40x_1x_2 + 10x_1x_2 + 20x_1x_3$$

Our problem can now be stated:

$$\text{Min } f(x_1, x_2, x_3) = 40/(x_1x_2x_3) + 40x_1x_2 + 10x_1x_2 + 20x_1x_3$$

s.t.:

$$400/(x_1x_2x_3) \leq 300$$

$$x_1, x_2, x_3 > 0$$



10

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Nonlinear Programming

Example

A close cylindrical tank is being designed to carry at least 20 m³ of chemicals. Metal for the top and sides cost \$50/m², but the heavier metal of the base cost \$200/m². Also, the height of the tank can be no more than twice its diameter to keep it from being top heavy. Formulate a constrained nonlinear program to find a design of minimum cost.

Modelling: The decision variables the designer must choose are x_1 , the diameter of the tank, and x_2 , the height of the tank.

$$\text{Volume of tank} = \pi(x_1 / 2)^2 x_2$$

$$\text{Area of top (or bottom)} = \pi(x_1 / 2)^2$$

$$\text{Area of side} = \pi x_1 x_2$$

$$\text{Metal cost: } 50(\pi x_1 x_2 + \pi(x_1 / 2)^2) + 200(\pi(x_1 / 2)^2)$$



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Nonlinear Programming

Then we obtain the NLP

$$\text{Min } f(x_1, x_2) = 50(\pi x_1 x_2 + \pi(x_1 / 2)^2) + 200(\pi(x_1 / 2)^2)$$

Subject to:

$$\pi(x_1 / 2)^2 x_2 \geq 20 \text{ (volume)}$$

$$x_2 \leq 2x_1 \text{ (height-to-diameter ratio)}$$

$$x_1 \geq 0, x_2 \geq 0$$



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Nonlinear Programming

• Single-variable Optimization

Single-variable optimization techniques are discussed first due to their simplicity. They provide a good understanding of the properties of the maximum and minimum points in a function and how optimization algorithms work iteratively to find the optimum point in a problem.

A single-variable, unconstrained, nonlinear program has the form

$$\text{Optimize: } z = f(x)$$

The search for the optimum is conducted over the infinite interval $(-\infty, \infty)$. If the search is restricted to a finite subinterval $[a, b]$, then the problem becomes

$$\text{Optimize: } z = f(x)$$

$$\text{Subject to: } a \leq x \leq b$$

which is a one-variable, constrained program.



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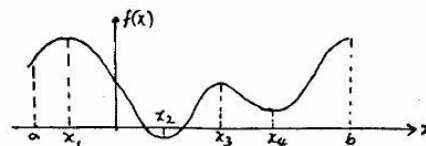
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Nonlinear Programming

Local and global optima

An objective function $f(x)$ has a *local minimum* at x_0 if there exists a (small) interval centered at x_0 such that $f(x) \geq f(x_0)$ for all x in this interval at which the function is defined. If $f(x) \geq f(x_0)$ for all x at which the function is defined, then the minimum at x_0 (besides being local) is a *global minimum*. Local and global maxima are defined similarly, in terms of $\leq$.

Consider the function defined only on $[a, b]$ as shown in Fig. 1.



It has local minima at a , x_2 , and x_4 ; local maxima at x_1 , x_3 , and b ; a global minimum at x_2 ; and global maxima at x_1 and b .



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Nonlinear Programming

The purpose of an optimization algorithm is to find a solution x , for which the function $f(x)$ is optimized. Locating optima by calculus is seldom useful in practice as in a lot of cases it is very difficult to solve the NLP analytically. Numerical methods are used to approximate the location of (some) local optima to within an acceptable tolerance.

Direct search methods

They use only objective function values to locate the optimum point.

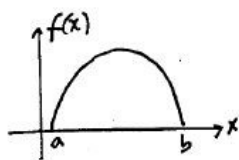
They start with a finite interval in which the objective function is presumed *unimodal*, that is, the interval is presumed to include one and only one point at which $f(x)$ has a local maximum or minimum (see Fig. 2)



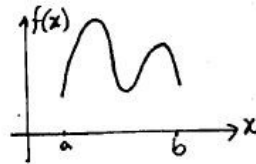
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Nonlinear Programming



A unimodal function
on $[a, b]$



A function that is
not unimodal on $[a, b]$

The techniques then systematically shrink the interval around the local optimum until the optimum is confined to within acceptable limits; the shrinking is effected by sequentially evaluating the objective function at selected points and then using the unimodal property to eliminate portions of the current interval.



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Nonlinear Programming

The fundamental rule for region-elimination method is as follows:

Consider two points x_1 and x_2 which lie in the interval $[a, b]$ and satisfy $x_1 < x_2$. For *unimodal* functions for minimization, we can conclude the following:

1. If $f(x_1) > f(x_2)$ then the minimum does not lie in $[a, x_1]$.
2. If $f(x_1) < f(x_2)$ then the minimum does not lie in $[x_2, b]$.
3. If $f(x_1) = f(x_2)$ then the minimum does not lie in $[a, x_1]$ and $[x_2, b]$.

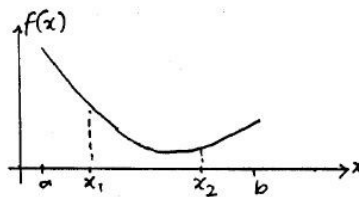


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Nonlinear Programming

Consider the unimodal function shown in Fig. 3



If $f(x_1) > f(x_2)$, the minimum point x^* cannot lie on the left side of x_1 . Thus, we can eliminate the region $[a, x_1]$ from further consideration. Therefore, we reduce our interval of interest from $[a, b]$ to $[x_1, b]$. Similarly, the second possibility ($f(x_1) < f(x_2)$) can be explained. If the third situation occurs ($f(x_1) = f(x_2)$), we can conclude that regions $[a, x_1]$ and $[x_2, b]$ can be eliminated with



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Nonlinear Programming

the assumption that there exists only one local minimum in the search space $[a, b]$.

The following algorithms constitute their search by using the fundamental rule for region elimination.

Fibonacci search method

In this method, the search number is reduced to Fibonacci numbers. The property of the Fibonacci numbers is that, given two consecutive numbers F_{n-2} and F_{n-1} , the third number is found as follows:

$$F_n = F_{n-1} + F_{n-2}, n = 2, 3, \dots$$

where $F_0 = F_1 = 1$. Thus, the Fibonacci sequence is $\{F_0, F_1, F_2, F_3, F_4, F_5, F_6, F_7, F_8, \dots\} = \{1, 1, 2, 3, 5, 8, 13, 21, 34\}$.

The Fibonacci search is initialized by finding the smallest

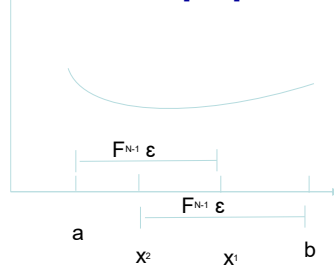


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Nonlinear Programming

Fibonacci number that satisfies $F_N \epsilon \geq b-a$, where ϵ is the prescribed tolerance and $[a, b]$ is the original interval of interest.



Set $\epsilon' \geq (b-a)/F_N$. The first two points in the search are located at $F_{N-1} \epsilon'$ from the endpoints of $[a, b]$. Successive points in the search are considered one at a time and are positioned at $F_j \epsilon'$ ($j = N-2, N-3, \dots, 2$) from the newest endpoint of the current interval.



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Nonlinear Programming

Example

Use the Fibonacci search to approximate the location of the maximum of $f(x) = x(5\pi - x)$ on $[0, 20]$ to within $\epsilon = 1$.

Initial points

Find the smallest Fibonacci number that satisfies $F_N \epsilon \geq b-a$.

The first Fibonacci number that satisfies

$$F_N \epsilon \geq 20-0$$

is $F_7 = 21$. We set $N = 7$,

$$\epsilon' = (b-a)/F_N = (20-0)/21 = 0.9524$$

and then position the first two points in the search at

$$F_{N-1} \epsilon' = F_6(0.9524) = 12.34$$

from each endpoint. Thus

$$x_1 = 0 + 12.38 = 12.38$$

$$x_2 = 20 - 12.38 = 7.62$$

$$f(x_1) = (12.38)(5\pi - 12.38) = 41.20$$

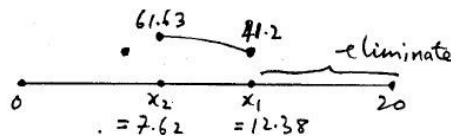
$$f(x_2) = (7.62)(5\pi - 7.62) = 61.63$$



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Nonlinear Programming



Using the unimodal property, we conclude that the maximum must occur to the left of 12.38, and we reduce the interval of interest to $[0, 12.38]$



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Nonlinear Programming

1st iteration

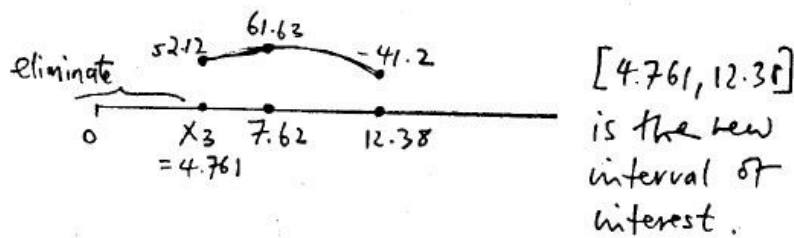
The next lower Fibonacci number (F_6 was the last one used) is $F_5 = 8$; so the next point in the search is positioned at

$$F_{N-1} \epsilon' = F_5(0.9524) = 7.619$$

from the newest endpoint, 12.38. Thus

$$x_3 = 12.38 - 7.619 = 4.761$$

$$f(x_3) = (4.761)(5\pi - 4.761) = 52.12$$



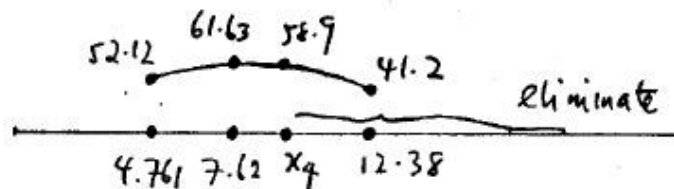
Nonlinear Programming

2nd iteration

The next lower Fibonacci number now is $F_4 = 5$. Thus,

$$x_4 = 4.761 + F_4 \epsilon' = 4.761 + 4(0.9524) = 9.523$$

$$F(x_4) = (9.523)(5\pi - 9.523) = 58.9$$



[4.761, 9.523] is the new interval of interest.

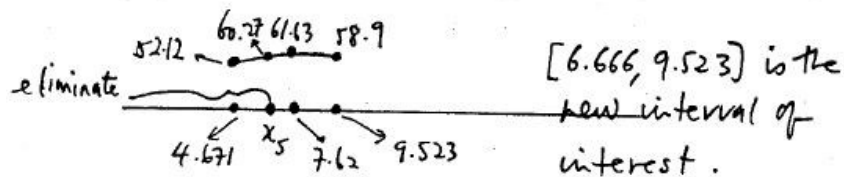
Nonlinear Programming

3rd iteration

The next lower Fibonacci number now is $F_3 = 3$, Hence

$$x_5 = 9.523 - F_3 \epsilon' = 9.523 - 3(0.9524) = 6.666$$

$$f(x_5) = (6.666)(5\pi - 6.666) = 60.27$$



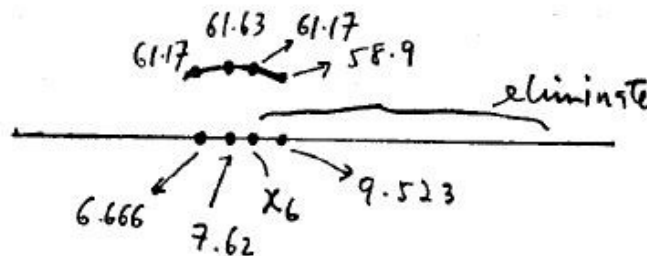
Nonlinear Programming

4th iteration

The next lower Fibonacci number now is $F_2 = 2$. Hence

$$x_6 = 6.666 + F_2 \epsilon' = 6.666 + 2(0.9524) = 8.571$$

$$F(x_6) = (8.571)(5\pi - 8.571) = 61.17$$



[6.666, 8.571] is the new interval of interest.

Nonlinear Programming

Note that $x_2 = 7.62$ is within $\epsilon = 1$ of every other point of the interval. We therefore accept x_2 as the location of the maximum, i.e.

$$x^* = x_2 = 7.62 \text{ with } z^* = f(x_2) = 61.63$$



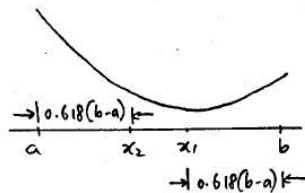
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Nonlinear Programming

Golden-section search method

In this algorithm, the search is based on the number 0.618, known as the golden section. The first two points of the search are located at $0.618(b-a)$ from the endpoints of the initial interval $[a, b]$. Successive points are considered one at a time and are positioned at $0.618L_i$ from the newest endpoint of the current interval, where L_i denotes the length of this interval.



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Nonlinear Programming

Example

Use the golden search method to approximate the location of the maximum of $f(x) = x(5\pi - x)$ on $[0, 20]$ to within $\epsilon = 1$.

The length of the initial interval is $L_1 = 20$, so the first two points in the search are positioned at $(0.618)(20) = 12.36$ from each endpoint. Thus,

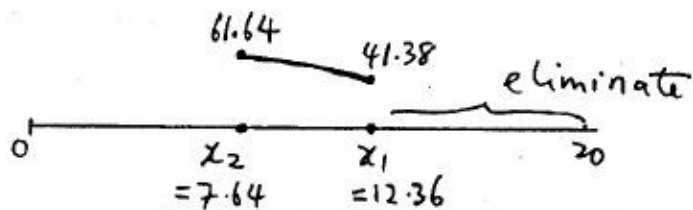
$$x_1 = 0 + 12.36 = 12.36$$

$$x_2 = 20 - 12.36 = 7.64$$

$$f(x_1) = (12.36)(5\pi - 12.36) = 41.38$$

$$f(x_2) = (7.64)(5\pi - 7.64) = 61.64$$

Nonlinear Programming



It follows from the unimodal property that the maximum must occur to the left of 12.36; hence $[0, 12.36]$ is the new interval of interest.

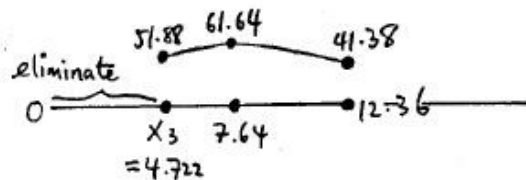
Nonlinear Programming

1st iteration

The interval has length $L_2 = 12.36$, so the next point in the search is positioned at $0.618L_2$ from the newest endpoint at 12.36. Therefore

$$x_3 = 12.36 - (0.618)(12.36) = 4.722$$

$$f(x_3) = (4.722)(5\pi - 4.722) = 51.88$$



$[4.722, 12.36]$ is the new interval of interest.

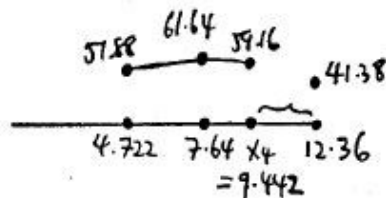
Nonlinear Programming

2nd iteration

$$L_3 = 12.36 - 4.722 = 7.638$$

$$x_4 = 4.722 + (0.618)(7.638) = 9.442$$

$$f(x_4) = (9.442)(5\pi - 9.442) = 59.16$$



$[4.722, 9.442]$
is the new
interval of
interest.

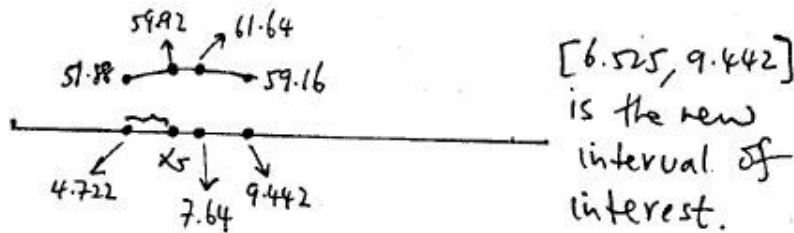
Nonlinear Programming

3rd iteration

$$L_4 = 9.442 - 4.722 = 4.720$$

$$x_5 = 9.442 - (0.618)(4.72) = 6.525$$

$$f(x_5) = (6.525)(5\pi - 6.525) = 59.92$$



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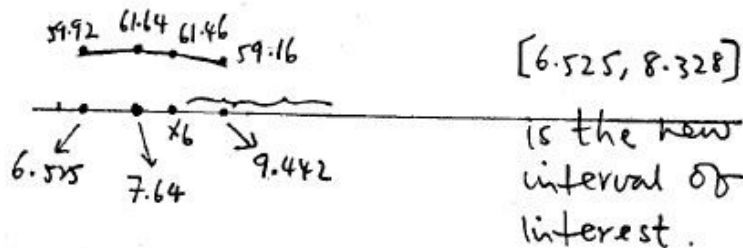
Nonlinear Programming

4th iteration

$$L_5 = 9.442 - 6.525 = 2.917$$

$$x_6 = 6.525 + (0.618)(2.917) = 8.328$$

$$f(x_6) = (8.328)(5\pi - 8.328) = 61.46$$



34

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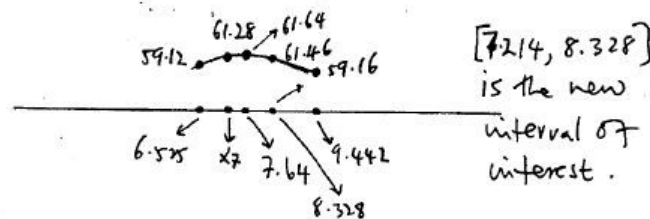
Nonlinear Programming

5th iteration

$$L_6 = 8.328 - 6.525 = 1.803$$

$$x_7 = 8.328 - (0.618)(1.803) = 7.214$$

$$f(x_7) = (7.214)(5\pi - 7.214) = 61.28$$



Note, however, the interval point $x_2 = 7.64$ is within $\epsilon = 1$ of all other points in the interval. So we take it as the location of the maximum, i.e.,

$$X^* = x_2 = 7.64 \text{ with } z^* = f(x_2) = 61.65$$

Nonlinear Programming

Gradient-based methods

Unlike the direct search methods, the gradient-based algorithms require derivative information.

Newton's method

In this method, a linear approximation to the first derivative of the function $f(x)$ is made at a point using the Taylor's series expansion:

$$f'(x) = f'(x_n) + f''(x_n)(x - x_n)$$

The point x_{n+1} is taken to be the point where the derivative of f is equal to 0. This gives $f'(x_n) + f''(x_n)(x_{n+1} - x_n)$ so that

$$x_{n+1} = x_n - f'(x_n)/f''(x_n) \quad n = 2, 1, \dots$$

where x_0 is an initial guess at a local optimum for f . The procedure is terminated when $|x_{n+1} - x_n| < \epsilon$ or when $|f'(x_{n+1})| < \epsilon$. Convergence of the algorithm depends on the initial point x_0 .

Nonlinear Programming

Example

Using Newton's method, find the minimum of $f(x) = x^2 + 54/x$.
The initial value is $x_0 = 2$. Stop when $|x_{n+1} - x_n| < 10^{-5}$.

We have

$$f'(x) = 2x - 54/x^2$$

$$f''(x) = 2 + 108/x^3$$

Thus,

$$\begin{aligned} x_1 &= x_0 - f'(x_0) / f''(x_0) \\ &= 2 - (2(2) - 54/2^2) / (2 + 108/2^3) \\ &= 2.6129 \end{aligned}$$



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Nonlinear Programming

The iterations are shown below

n	x_n	x_{n+1}	$ x_{n+1} - x_n $
0	2	2.612903	0.612903
1	2.612930	2.946104	0.333201
2	2.946104	2.99902	0.0529154
3	2.999020	2.999999	0.000979
4	2.99999	2.999999	0.000000

Take $x^* = x_5 = 2.99999 \approx 3.0$



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Nonlinear Programming

• Multivariable Optimization

Consider the following problem:

Optimize: $z = f(X)$, where $X = [x_1, x_2, \dots, x_n]^T$.

We shall assume that the optimization to be a minimization;

All results will apply to a maximization program if $f(X)$ is replaced by $-f(X)$.

Gradient vector and Hessian matrix

The gradient vector at any point $\hat{X}$ is represented by $\nabla f(\hat{X})$ which is an n-dimensional vector given as follows:

$$\nabla f(\hat{X}) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right)^T$$

The Hessian matrix associated with a function $f(X)$ is represented by $\nabla^2 f(X)$. It is a matrix with entries

$$[\nabla^2 f(X)]_{ij} = \frac{\partial^2 f(X)}{\partial x_i \partial x_j}$$



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Nonlinear Programming

Example

Consider the following function

$$f(x_1, x_2) = x_1^3 + 2x_1x_2 + x_2^2$$

which has

$$\nabla f(x_1, x_2) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 3x_1^2 + 2x_2 \\ 2x_1 + 2x_2 \end{bmatrix}$$

and

$$\nabla^2 f(x_1, x_2) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 6x_1 & 2 \\ 2 & 2 \end{bmatrix}$$



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Nonlinear Programming

Example

Consider the function

$$f(x_1, x_2) = 2x_1^4 + 3x_1^2x_2 + 2x_1x_2^3 + 4x_2^2$$

The gradient of this function is

$$\nabla f(x) = \begin{bmatrix} 8x_1^3 + 6x_1x_2 + 2x_2^3 \\ 3x_1^2 + 6x_1x_2^2 + 8x_2 \end{bmatrix}$$

and the Hessian matrix is

$$\nabla^2 f(x) = \begin{bmatrix} 24x_1^2 + 6x_2 & 6x_1 + 6x_2^2 \\ 6x_1 + 6x_2^2 & 12x_1x_2 + 8 \end{bmatrix}$$

At the point $X_0 = (-2, 3)^T$ these become

$$\nabla f(x_0) = \begin{bmatrix} -46 \\ -72 \end{bmatrix} \quad \nabla^2 f(x_0) = \begin{bmatrix} 114 & 42 \\ 42 & -64 \end{bmatrix}$$



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Nonlinear Programming

Optimality criteria

A point X^* is a *stationary point* if $\nabla f(X^*) = 0$. Furthermore, the point is a *minimum* or a *maximum*, if $\nabla^2 f(X^*)$ is *positive-definite* or *negative-definite*.

Definition

An $n \times n$ symmetric matrix A (such that $A = A^T$) is positive (or negative)-definite if $X^T A X > 0$ (or < 0) for every $X \neq 0$. It is said to be indefinite if $X^T A X > 0$ for some x and $X^T A X < 0$ for other x .

Test for positive-definiteness

Let $A = [a_{ij}]$ be an $n \times n$ symmetric matrix, and define the determinants

$$A_1 = |a_{11}|, A_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, A_3 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \dots, A_n = |A|$$



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Nonlinear Programming

Then A is *positive-definite* if and only if $A_1 > 0, A_2 > 0, \dots, A_n > 0$.

A is positive *semi-definite* if $A_1 \geq 0, A_2 \geq 0, \dots, A_n \geq 0$.

Note that $A_1, A_2, \dots, A_n$ are called the *principal minors* of A .

A is negative-definite if and only if $A_1 < 0, A_2 > 0, A_3 < 0, \dots$, (note the alternating signs).

Example

(a) $H = \begin{bmatrix} 2/\sqrt{5} & 2/5 \\ 2/5 & 12/(5\sqrt{5}) \end{bmatrix}$ is positive-definite since

$$H_1 = |2/\sqrt{5}| = 2/\sqrt{5} > 0$$

$$H_2 = \begin{vmatrix} 2/\sqrt{5} & 2/5 \\ 2/5 & 12/(5\sqrt{5}) \end{vmatrix} = 20/25 > 0$$



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Nonlinear Programming

(b) $H = \begin{bmatrix} -8 & 0 \\ 0 & -30 \end{bmatrix}$ is negative-definite since

$$H_1 = |-8| = -8 < 0$$

$$H_2 = \begin{vmatrix} -8 & 0 \\ 0 & -30 \end{vmatrix} = 240 > 0$$

(c) $H = \begin{bmatrix} 2 & 0 & 1 \\ 0 & -3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ is indefinite since

$$H_1 = 2 > 0, H_2 = -6 < 0, H_3 = -9 < 0$$



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Nonlinear Programming

Example

Consider the function

$$f(x_1, x_2) = x_1 (70 - 4x_1) + x_2 (150 - 15x_2) - 100 - 15x_1 - 15x_2$$

To find the stationary points for $f(x_1, x_2)$, let

$$\frac{\partial f}{\partial x_1} = 70 - 8x_1 - 15 = 0 \text{ (for } x_1 = 55/8 \text{)}$$

$$\frac{\partial f}{\partial x_2} = 150 - 30x_2 - 15 = 0 \text{ (for } x_2 = 9/2 \text{)}$$

Thus, the only stationary point of $f(x_1, x_2)$ is $(55/8, 9/2)^T$. The

Hessian matrix for $f(x_1, x_2)$ is

$$H(x_1, x_2) = \begin{bmatrix} -8 & 0 \\ 0 & -30 \end{bmatrix}$$

We have

$$H_1 = |-8| = -8 < 0 \quad H_2 = \begin{vmatrix} -8 & 0 \\ 0 & -30 \end{vmatrix} = 240 > 0$$

Thus, H is negative definite and $(55/8, 9/2)^T$ is a local maximum.



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Nonlinear Programming

Example

Consider the function

$$f(x_1, x_2) = 2x_1 + x_1^{-2} + 2x_2 + x_2^{-2}$$

Which has

$$\nabla f = \begin{pmatrix} 2 - 2x_1^{-3} \\ 2 - 2x_2^{-3} \end{pmatrix}$$

$$H = \begin{pmatrix} 6x_1^{-4} & 0 \\ 0 & 6x_2^{-4} \end{pmatrix}$$

And a stationary point $x^* = (1 \ 1)^T$

$$\text{At } x^* = (1 \ 1)^T, H = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix} > 0$$

Thus, $x^* = (1 \ 1)^T$ is a local minimum.



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Nonlinear Programming

Example

Consider the function

$$f(x_1, x_2) = 1/3x_1^3 + 1/2x_1^2 + 2x_1x_2 + 1/2x_2^2 - x_2 + 9$$

The condition for stationary point is

$$\nabla f(x) = \begin{bmatrix} x_1^2 + x_1 + 2x_2 \\ 2x_1 + x_2 - 1 \end{bmatrix} = 0$$

The 2nd component of this condition shows that $x_2 = 1 - 2x_1$, and if this is used in the 1st component we get

$$x_1^2 - 3x_1 + 2 = 0 \text{ or } (x_1 - 1)(x_1 - 2) = 0$$

Hence there are 2 stationary points:

$$x_a = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ and } x_b = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

The Hessian matrix for the function is

$$\nabla^2 f(x) = \begin{bmatrix} 2x_1 + 1 & 2 \\ 2 & 1 \end{bmatrix}$$



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Nonlinear Programming

$$\nabla^2 f(x_a) = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} \text{ and } \nabla^2 f(x_b) = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$$

$\nabla^2 f(x_b)$ is positive definite, so x_b is a local min. However, $\nabla^2 f(x_a)$ is indefinite, and x_a is neither a min or max of f .

The function f has neither a global min or max, since f is unbounded as $x_1 \rightarrow \pm \infty$.



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Nonlinear Programming

Example

There are two firms producing widgets. It costs the first firm $\$x_1$ to produce x_1 widgets, and it costs the second firm $\$0.5x_2^2$ to produce x_2 widgets. If a total of x widgets are produced, consumers will pay $\$(200 - x)$ for each widgets. If the two manufacturers want to collude in an attempt to maximize the sum of their profits, how many widgets should each company produce?



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Nonlinear Programming

$$\begin{aligned} \text{Profit} &= \text{revenue} - \text{cost of production} \\ &= (200 - x)x - (x_1 + 0.5x_2^2) \end{aligned}$$

The NLP is:

$$\text{Max } f = (200 - x)x - (x_1 + 0.5x_2^2)$$

$$\text{s.t. } x_1 + x_2 = x$$

- Convert this constrained problem into an unconstrained problem.
- Substitute the constraint into the objective function gives

$$\begin{aligned} f &= (200 - x_1 - x_2)(x_1 + x_2) - x_1 - 0.5x_2^2 \\ &= 199x_1 + 200x_2 - x_1^2 - 2x_1x_2 - 1.5x_2^2 \end{aligned}$$

The condition for stationary points is

$$\nabla f(x) = \begin{bmatrix} 199 - 2x_1 - 2x_2 \\ 200 - 2x_1 - 3x_2 \end{bmatrix} = 0$$



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Nonlinear Programming

Solving the 2 simultaneous equations give $x_1 = 98.5$ and $x_2 = 1$.

The Hessian matrix for the function f is

$$\nabla^2 f(x) = \begin{bmatrix} -2 & -2 \\ -2 & -3 \end{bmatrix}$$

which is negative definite and $(x_1, x_2) = (98.5, 1)$ is a local max.



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Nonlinear Programming

Newton's method

In some cases, it is easy to find a stationary point. But in many problems, it may be difficult. So numerical methods are used to approximate (local) minima to within prescribed tolerances.

Choose an initial vector X_0 . Vectors $X_1, X_2, X_3, \dots$ then are determined iteratively by

$$X_{n+1} = X_n - [\nabla^2 f(X_n)]^{-1} \nabla f(X_n)$$

The iteration process terminates if and when $|f(X_n) - f(X_{n+1})| < \epsilon$, where ϵ is the prescribed tolerance. The last computed X -vector becomes the final approximation to the *local minimum* X^* . The method will converge to a local minimum if $\nabla^2 f(X_n)$ is *positive-definite* in the vicinity of X^* and X_0 is close to X^* .



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Nonlinear Programming

Example

Use the Newton method to

Maximize: $z = -(x_1 - \sqrt{5})^2 - (x_2 - \pi)^2 - 10$

to within a tolerance of $\epsilon = 0.05$. Take $X_0 = [6.597, 5.891]^T$, with $f(X_0) = -36.58$.

The gradient vector, Hessian matrix and inverse Hessian matrix are

$$\nabla f(x) = \begin{bmatrix} -2(x_1 - \sqrt{5}) \\ -2(x_2 - \pi) \end{bmatrix}, H = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}, H^{-1} = \begin{bmatrix} -0.5 & 0 \\ 0 & -0.5 \end{bmatrix}$$

for all x_1 and x_2 .



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Nonlinear Programming

First iteration

$$\nabla f(x_0) = \begin{bmatrix} -2(6.597 - \sqrt{5}) \\ -2(5.891 - \pi) \end{bmatrix} = \begin{bmatrix} -8.722 \\ -5.499 \end{bmatrix}$$

$$X_1 = X_0 - H(X_0)^{-1} \nabla f(x_0)$$

$$= \begin{bmatrix} 6.597 \\ 5.891 \end{bmatrix} - \begin{bmatrix} -0.5 & 0 \\ 0 & -0.5 \end{bmatrix} \begin{bmatrix} -8.722 \\ -5.499 \end{bmatrix} = \begin{bmatrix} 2.236 \\ 3.142 \end{bmatrix}$$

with $f(x_1) = -10.00$. Since

$$\begin{aligned} f(X_1) - f(X_0) &= -10.00 - (-36.58) \\ &= 26.58 > 0.05 \end{aligned}$$

We continue iterating.



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Nonlinear Programming

2nd iteration

$$\nabla f(X_1) = \begin{bmatrix} -2(2.236 - \sqrt{5}) \\ -2(3.142 - \pi) \end{bmatrix} = \begin{bmatrix} -0.0001 \\ 0.0008 \end{bmatrix}$$

$$\begin{aligned} X_2 &= X_1 - H(X_1)^{-1} \nabla f(x_1) \\ &= \begin{bmatrix} 2.236 \\ 3.142 \end{bmatrix} - \begin{bmatrix} -0.5 & 0 \\ 0 & -0.5 \end{bmatrix} \begin{bmatrix} -0.0001 \\ -0.0008 \end{bmatrix} = \begin{bmatrix} 2.236 \\ 3.142 \end{bmatrix} \end{aligned}$$

with $f(x_2) = -10.00$. Since

$f(X_2) - f(X_1) = 0 < 0.05$, we take $X^* = X_2 = [2.236, 3.142]^T$

with $z^* = f(x_2) = -10.00$.



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Nonlinear Programming

Example

Use the Newton method to

Maximize: $z = -\sin(x_1 x_2) + \cos(x_1 - x_2)$

to within a tolerance of $\epsilon = 0.05$, starting with $X_0 = [4.8, 1.6]^T$,
with $f(X_0) = -1.983$.

The gradient vector, Hessian matrix and inverse Hessian matrix
are

$$\nabla f(x) = \begin{bmatrix} -x_2 \cos(x_1 x_2) - \sin(x_1 - x_2) \\ -x_1 \cos(x_1 x_2) + \sin(x_1 - x_2) \end{bmatrix},$$

$$H = \begin{bmatrix} x_2^2 \sin(x_1 x_2) - \cos(x_1 - x_2) & -\cos(x_1 x_2) + x_1 x_2 \sin(x_1 x_2) + \cos(x_1 - x_2) \\ -\cos(x_1 x_2) + x_1 x_2 \sin(x_1 x_2) + \cos(x_1 - x_2) & x_2^2 \sin(x_1 x_2) - \cos(x_1 - x_2) \end{bmatrix}$$



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Nonlinear Programming

First iteration

$$\nabla f(x_0) = \begin{bmatrix} -0.2186 \\ -0.8893 \end{bmatrix}, H(x_0) = \begin{bmatrix} 3.52 & 6.393 \\ 6.393 & 23.69 \end{bmatrix},$$

$$H(x_0)^{-1} = \begin{bmatrix} 0.5572 & -0.1504 \\ -0.1504 & 0.08279 \end{bmatrix}$$

Then

$$\begin{aligned} X_1 &= X_0 - H(x_0)^{-1} \nabla f(x_0) \\ &= \begin{bmatrix} 4.788 \\ 1.641 \end{bmatrix} \end{aligned}$$

with $f(x_1) = -2.0$. Even though x_1 is close to x_0 , we have

$$f(x_1) < f(x_0)$$

and the iterations are tending toward a minimum rather than a maximum

Note that $H(x_0)$ is not negative-definite; in fact, it is positive-definite.

A different value for x_0 must be used if the Newton method is to succeed.



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Nonlinear Programming

- **Multivariable optimization with equality constraints**

Standard forms

with $X = [x_1, x_2, \dots, x_n]^T$, standard form for NL programs containing only *equality constraints* is

$$\text{Minimize: } z = f(X) \quad (1)$$

$$\text{Subject to: } h_i(X) = 0, i = 1, 2, \dots, m$$

where $m < n$ (fewer constraints than variables).

Maximization programs are converted into minimization programs by multiplying the objective function by -1.

To solve programs in which all the constraints are equality (or inequality) constraints, *Lagrange multipliers* can be used.



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Nonlinear Programming

Lagrange multipliers

The method is equivalent to using the constraint equations to eliminate certain of the x -variables from the objective function and then solving an unconstrained minimization program in the remaining x -variables. To solve program (1), first form the Lagrangian function

$$L(X, \lambda) = f(X) + \sum_{i=1}^m \lambda_i h_i(X)$$

where λ_i ($i = 1, 2, \dots, m$) are (unknown) constants called *Lagrange multipliers*. Then solve the system of $n+m$ equations

$$\frac{\partial L}{\partial x_j} = 0 \quad (j = 1, 2, \dots, n) \text{ or } \nabla_x L = 0$$

$$\frac{\partial L}{\partial \lambda_i} = 0 \quad (i = 1, 2, \dots, m) \text{ or } \nabla_\lambda L = 0 \quad (2)$$

The *sufficient condition* for a local minimum of f are given in the theorem below. That is, the theorem states that if the condition are satisfied, then X^* is a local minimum of f .



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Nonlinear Programming

Theorem (sufficient conditions)

Let X^* be a point satisfying $h(X^*) = 0$, and a vector $\lambda^* = [\lambda_1^*, \lambda_2^*, \dots, \lambda_m^*]^T$ such that

$$\nabla_x L(X^*, \lambda^*) = 0.$$

Suppose also the matrix $\nabla_{xx}^2 L(X^*, \lambda^*)$ is *positive definite* on $M = \{Y: \nabla h(X^*)^T Y = 0\}$, that is, for $Y \in M$, $Y \neq 0$ there holds

$Y^T \nabla_{xx}^2 L(X^*, \lambda^*) Y > 0$. Then X^* is a strict local minimum of f subject to $h(X) = 0$.

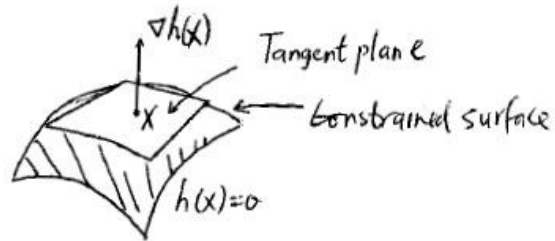
The set $M = \{Y: \nabla h(X^*)^T Y = 0\}$ is the *tangent space* at a point X^* on the surface $S = \{X: h(X) = 0\}$. The above theorem states that if a point X^* satisfies $\nabla_x L(X^*, \lambda^*) = 0$, and $\nabla_{xx}^2 L(X^*, \lambda^*)$ is positive-definite on M , then X^* is a strict local minimum of f .



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Note that some assumptions must be made to ensure the validity of the optimality conditions. We have chosen to assume that a solution X^* to the optimization program is a *regular* point. In the case of equality constraints, this means that the gradients of the constraints $\{\nabla h_i(X^*)\}$ are linearly independent.



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Nonlinear Programming

Example

Let $S = \{X \in \mathbb{R}^3 : h_1(X) = x_1 = 0, h_2(X) = x_1 - x_2 = 0\}$. S is the x_3 axis in $\mathbb{R}^3$. Then

$$\nabla h_1(X) = [1 \ 0 \ 0]^T$$

$$\nabla h_2(X) = [1 \ -1 \ 0]^T$$

Hence, the vectors $\nabla h_1(X)$ and $\nabla h_2(X)$ are linearly independent evaluated at any $X \in S$. That is, the matrix given by

$$\begin{bmatrix} \nabla h_1(X)^T \\ \nabla h_2(X)^T \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \end{bmatrix}$$

has $\text{rank} = m = 2$.

The tangent space at an arbitrary point of S is

$$M = \{Y : \nabla h_1(X)^T Y = 0, \nabla h_2(X)^T Y = 0\}$$



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Nonlinear Programming

Thus,

$$M = \{Y: \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}\}$$

$$= \{[0 \ 0 \ \alpha]\}$$

$$= \text{the } x_3 \text{ axis in } \mathbb{R}^3$$

Example

Suppose at $X^* = 0$, $\nabla h_1(X^*) = [-2 \ 0]^T$, $\nabla h_2(X^*) = [-4 \ 0]^T$.
Then $\nabla h_1(X^*)$ and $\nabla h_2(X^*)$ are linearly dependent. The matrix give by

$$\begin{bmatrix} \nabla h_1(X^*)^T \\ \nabla h_2(X^*)^T \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ -4 & 0 \end{bmatrix}$$

has rank $1 < 2$.



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Nonlinear Programming

Example

A company is planning to spend \$10K on advertising. It costs \$3K per minute to advertise on TV and \$1K per minute to advertise on radio. If the firm buys x_1 minutes of TV advertising and x_2 minutes of radio advertising, its revenue in \$K is given by

$$f(x_1, x_2) = -2x_1^2 - x_2^2 + x_1x_2 + 8x_1 + 3x_2$$

How can the firm maximize its revenue?

We wish to solve the following NLP:

$$\text{Max: } z = -2x_1^2 - x_2^2 + x_1x_2 + 8x_1 + 3x_2$$

$$\text{Subject to: } 3x_1 + x_2 = 10$$



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Nonlinear Programming

Applying the method of Lagrange multipliers to the original program standardized as

$$\text{Minimize: } z = 2x_1^2 + x_2^2 - x_1x_2 - 8x_1 - 3x_2$$

$$\text{Subject to: } 3x_1 + x_2 = 10$$

Here,

$$f(x_1, x_2) = 2x_1^2 + x_2^2 - x_1x_2 - 8x_1 - 3x_2$$

$$h(x_1, x_2) = 3x_1 + x_2 - 10$$

The Lagrange function is then:

$$L = f(X) + \lambda h(X)$$

$$= 2x_1^2 + x_2^2 - x_1x_2 - 8x_1 - 3x_2 + \lambda(3x_1 + x_2 - 10)$$



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Nonlinear Programming

We have

$$\frac{\partial L}{\partial x_1} = 4x_1 - x_2 - 8 + 3\lambda = 0 \quad (1)$$

$$\frac{\partial L}{\partial x_2} = 2x_2 - x_1 - 3 + \lambda = 0 \quad (2)$$

$$\frac{\partial L}{\partial \lambda} = 3x_1 + x_2 - 10 = 0 \quad (3)$$

We have to solve equations (1) – (3)

From (1) and (2), we obtain

$$-7x_1 + 7x_2 - 1 = 0 \quad (4)$$



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Nonlinear Programming

From (3) and (4), we obtain

$$x_1 = 69/28, x_2 = 73/28$$

Using these in (1) gives $\lambda = 1/4$.

Is $X = (69/28, 73/28)^T$ a minimum point?

Now, we want to find $M = \{Y: \nabla h(X)^T Y = 0\}$
 $\nabla h(X) = [3 \ 1]^T$

and

$$\begin{aligned} \nabla h(X)^T Y &= [3 \ 1] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \\ &= 3y_1 + y_2 = 0 \end{aligned} \quad (5)$$



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Nonlinear Programming

That is, $M = \{Y: 3y_1 + y_2 = 0\}$

We note that

$$\nabla_{xx}^2 L = \begin{bmatrix} \frac{\partial^2 L}{\partial x_1^2} & \frac{\partial^2 L}{\partial x_1 \partial x_2} \\ \frac{\partial^2 L}{\partial x_2 \partial x_1} & \frac{\partial^2 L}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ -1 & 2 \end{bmatrix}$$

and on M ,

$$\begin{aligned} Y^T \nabla_{xx}^2 L Y &= 4y_1^2 - 2y_1y_2 + 2y_2^2 \\ &= 4y_1^2 + 6y_1^2 + 2y_2^2 \text{ (using (5))} \\ &= 10y_1^2 + 2y_2^2 > 0 \quad \forall y \neq 0 \end{aligned}$$

$\Rightarrow X$ is a minimum point.

$\Rightarrow$ Thus, X is a maximum point of the original problem, with $z^* \approx \$15K$



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Nonlinear Programming

Example:

Consider the problem:

$$\text{Min } f(x_1, x_2) = (x_1 - 2)^2 + (x_2 - 2)^2$$

$$\text{s.t. } h(x_1, x_2) = x_1^2 + x_2^2 - 1 = 0$$

The Lagrangian function is

$$L = (x_1 - 2)^2 + (x_2 - 2)^2 + \lambda(x_1^2 + x_2^2 - 1)$$

We have

$$\frac{\partial L}{\partial x_1} = 2(x_1 - 2) + 2\lambda x_1 = 0 \quad (1)$$

$$\frac{\partial L}{\partial x_2} = 2(x_2 - 2) + 2\lambda x_2 = 0 \quad (2)$$

$$\frac{\partial L}{\partial \lambda} = x_1^2 + x_2^2 - 1 = 0 \quad (3)$$



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Nonlinear Programming

From (1) and (2), we get

$$2[(\lambda + 1)x_1 - 2] = 0 \quad (4)$$

$$2[(\lambda + 1)x_2 - 2] = 0 \quad (5)$$

(4) and (5) imply that $x_1 = x_2$. Thus, from (3), we get $x_1 = x_2 = 1/\sqrt{2}$ and from (4) or (5), we get $\lambda = 2\sqrt{2} - 1$.

Is $(x_1, x_2) = (1/\sqrt{2}, 1/\sqrt{2})$ a minimum point?

We want to find the subspace $M = \{Y: \nabla h(X)^T Y = 0\}$. Here,

$$\begin{aligned} \nabla h(X) &= [2x_1 \ 2x_2]^T \\ \nabla h(X)^T Y &= [2x_1 \ 2x_2] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \\ &= 2x_1 y_1 + 2x_2 y_2, \quad x_1 = x_2 \\ &= 2x_1(y_1 + y_2) \end{aligned} \quad (6)$$



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Nonlinear Programming

We note that

$$\nabla_{xx}^2 L = \begin{bmatrix} 2(1+\lambda) & 0 \\ 0 & 2(1+\lambda) \end{bmatrix}$$

On M,

$$Y^T \nabla_{xx}^2 L Y = 2(1+\lambda)(y_1^2 + y_2^2) > 0 \quad \forall Y \neq 0$$

$\Rightarrow (x_1, x_2) = (1/\sqrt{2}, 1/\sqrt{2})$ is a minimum point.



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Nonlinear Programming

Example:

A soft drinks company has divided a township into two territories. If x_1 dollars are spent on promotion in Zone A, then $6x_1^{1/2}$ cases of soft drink can be sold in zone A, and if x_2 dollars are spend on promotion in Zone B, then $4x_2^{1/2}$ cases of soft drinks can be sold in zone B. Each case of soft drinks sold in Zone A sells for \$10 and incurs\$5 in shipping and production costs. Each case of soft drinks sold in Zone B sells for \$9 and incurs \$4 in shipping and production costs. A total of \$100 is available for promotion. How can the soft drinks company maximize profits?

Profit = revenue – [cost of production + promotion + shipping costs]



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Nonlinear Programming

$$\begin{aligned} f_1 &= 10(6x_1^{1/2}) + 9(4x_2^{1/2}) - [x_1 + 5(6x_1^{1/2}) + x_2 + 4(4x_2^{1/2})] \\ &= 30x_1^{1/2} + 20x_2^{1/2} - x_1 - x_2 \end{aligned}$$

The NLP is:

$$\begin{aligned} &\text{Max } f_1 \\ &\text{s.t. } x_1 + x_2 = 100 \end{aligned}$$

We convert the problem to a min problem by letting $f = -f_1$.

The Lagrangian function is

$$\begin{aligned} L &= f(x) + \lambda h(x) \\ &= -30x_1^{1/2} - 20x_2^{1/2} + x_1 + x_2 + \lambda(x_1 + x_2 - 100) \end{aligned}$$



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Nonlinear Programming

We have

$$\frac{\partial L}{\partial x_1} = -15x_1^{-1/2} + 1 + \lambda = 0 \quad (1)$$

$$\frac{\partial L}{\partial x_2} = -10x_2^{-1/2} + 1 + \lambda = 0 \quad (2)$$

$$\frac{\partial L}{\partial \lambda} = x_1 + x_2 - 100 = 0 \quad (3)$$

From (1) and (2), we get

$$\frac{15}{x_1^{1/2}} = \frac{10}{x_2^{1/2}} \quad (4)$$

Squaring both sides and re-arranging gives

$$4x_1 - 9x_2 = 0 \quad (5)$$

From (3) and (5), we get

$$(x_1, x_2) = (900/13, 400/13)$$

Using this in (1) gives $\lambda = \frac{\sqrt{13}}{2} - 1$



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Nonlinear Programming

We want to find the subspace $M = \{Y: \nabla h(X)^T Y = 0\}$. Here

$$\nabla h(X) = [1 \ 1]^T$$

$$\begin{aligned} \nabla h(X)^T Y &= [1 \ 1] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \\ &= y_1 + y_2 = 0 \end{aligned} \quad (6)$$

We note that

$$\nabla_{xx}^2 L = \begin{bmatrix} 7.5x_1^{-3/2} & 0 \\ 0 & 5x_2^{-3/2} \end{bmatrix}$$

On M ,

$$Y^T \nabla_{xx}^2 L Y = 7.5x_1^{-3/2}y_1^2 + 5x_2^{-3/2}y_2^2 > 0 \quad \forall Y \neq 0$$

$$\Rightarrow (x_1, x_2) = (900/13, 400/13) \text{ is a min point.}$$

Hence, it is a max point of the original problem.



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Nonlinear Programming

• Optimization with inequality constraints

Consider programs of the form

Minimize: $f(X)$

Subject to: $g_j(X) \leq 0, j = 1, 2, \dots, p$

An inequality constraint $g_j \leq 0$ is said to be *active* at X if $g_j(X) = 0$. It is *inactive* at X if $g_j(X) < 0$.

Let the Lagrangian function to be

$$L(X, \mu) = f(X) + \sum_{j=1}^p \mu_j g_j(X)$$



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Nonlinear Programming

The sufficient conditions for a local minimum of f are given below

Theorem (sufficient condition)

A point X^* is a strict local *minimum* of f in the set $\{X: g(X) \leq 0\}$ if there exist μ^* such that

- (1) $\nabla_x L(X^*, \mu^*) = 0$
- (2) $g(X^*) \leq 0$
- (3) $\mu_j^* g_j(X^*) = 0, j = 1, 2, \dots, p$
- (4) $\mu^* \geq 0$
- (5) $Y^T \nabla_{xx}^2 L(X^*, \mu^*) Y > 0$ on $M' = \{Y: \nabla g_j(X^*)^T Y = 0 \text{ for all } j \in J\}$, where $J = \{j: g_j(X^*) = 0, \mu_j^* > 0\}$.



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Note:

Conditions (1) – (4) are the necessary conditions, sometimes known as the Kuhn-Tucker (K-T) conditions, for the program.

K-T points (i.e., points that satisfy the K-T conditions) are likely candidates for the optimum. However, not all K-T points are optimal points.

To check whether a K-T point is optimal, the additional condition (5) is used. J is the set of indices for which $g_j(X^*) = 0, \mu_j^* > 0$.

We have assumed that X^* is *regular*, that is, the gradients of the *active* constraints at X^* , $\{\nabla g_j(X^*): \nabla g_j(X^*) = 0, j \in J\}$, are linearly independent.



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Nonlinear Programming

Example: Solve the following nonlinear program:

$$\text{Min: } z = -x_1(30 - x_1) - x_2(50 - 2x_2) + 3x_1 + 5x_2 + 10x_3$$

$$\text{Subject to: } x_1 + x_2 - x_3 \leq 0$$

$$x_3 \leq 17.25$$

The Lagrange function is

$$\begin{aligned} L &= f(x) + (\mu_1 g_1 + \mu_2 g_2) \\ &= -x_1(30 - x_1) - x_2(50 - 2x_2) + 3x_1 + 5x_2 + 10x_3 + \\ &\quad \mu_1 (x_1 + x_2 - x_3) + \mu_2 (x_3 - 17.25) \end{aligned}$$



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Nonlinear Programming

We have

$$\frac{\partial L}{\partial x_1} = -30 + 2x_1 + 3 + \mu_1 = 0 \quad (1)$$

$$\frac{\partial L}{\partial x_2} = -50 + 4x_2 + 5 + \mu_1 = 0 \quad (2)$$

$$\frac{\partial L}{\partial x_3} = 10 - \mu_1 + \mu_2 = 0 \quad (3)$$

$$g_1 = x_1 + x_2 - x_3 \leq 0 \quad (4)$$

$$g_2 = x_3 - 17.25 \leq 0 \quad (5)$$

$$\mu_1 g_1 = \mu_1 (x_1 + x_2 - x_3) = 0 \quad (6)$$

$$\mu_2 g_2 = \mu_2 (x_3 - 17.25) = 0 \quad (7)$$

$$\mu_1 \geq 0, \mu_2 \geq 0 \quad (8)$$

We need to solve eqns. (1) – (8). This is a system of equalities and inequalities, requiring a case-by-case examination of the possible solutions. The 4 possibilities are as follows:



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- (a). $\mu_1 = \mu_2 = 0$. This cannot occur as eqn. (3) would be violated.
 (b). $\mu_1 = 0, \mu_2 > 0$. If $\mu_1 = 0$, then eqn. (3) implies $\mu_2 = -10 < 0$. This would violate eqn. (8).
 (c). $\mu_1 > 0, \mu_2 = 0$, and (d). $\mu_1 > 0, \mu_2 > 0$. Both would give the same optimal solution. Hence, only case(c) would be considered.

If $\mu_2 = 0$, then from eqn. (3), $\mu_1 = 10$. Now, from eqn. (1), $x_1 = 8.5$, and from eqn. (2), $x_2 = 8.75$. From eqn. (6), we have $x_1 + x_2 - x_3 = 0$. Solving for x_3 gives $x_3 = 17.25$. Thus, $X = (8.5, 8.75, 17.25)^T$, $\mu = (10, 0)^T$ satisfy the K-T condition. But is X a minimum point? That is, does it satisfy condition (5) of the theorem?

Note that $g_1(X) = g_2(X) = 0$. However, the active constraint $g_2(X) = 0$ does not enter the computation of $M' = \{Y: \nabla g_j(X_*)^T Y = 0, \text{ for all } j \in J\}$ since $\mu_2 = 0$.

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$$\nabla g_1^T Y = [1 \quad 1 \quad -1] \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = y_1 + y_2 - y_3 = 0$$

$$\nabla_{xx}^2 L(X, \mu) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \geq 0 \text{ on } \mathbb{R}^3. \text{ (Verify)}$$

On M' ,

$$\begin{aligned} Y^T \nabla_{xx}^2 L(X, \mu) Y &= [y_1 \quad y_2 \quad y_3] \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\ &= [y_1 \quad y_2 \quad y_3] \begin{bmatrix} 2y_1 \\ 4y_2 \\ 0 \end{bmatrix} \\ &= 2y_1^2 + 4y_2^2 > 0 \text{ for all } Y \neq 0. \end{aligned}$$

This implies that $\nabla_{xx}^2 L$ is positive definite on M' .

Thus, $X = (8.5, 8.75, 17.25)^T$ is a minimum point.

Nonlinear Programming

Consider programs of the form

Minimize: $f(X)$

Subject to: $h_i(X) = 0, i = 1, 2, \dots, m$

$g_j(X) \leq 0, j = 1, 2, \dots, p$

Let the Lagrangian function be

$$L(X, \lambda, \mu) = f(X) + \sum_{i=1}^m \lambda_i h_i(X) + \sum_{j=1}^p \mu_j g_j(X)$$

The sufficient conditions for a local minimum of f are given below.



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Theorem: A point X^* is a strict local *minimum* of f in the set $\{X: h(X) = 0, g(X) \leq 0\}$ if there exist λ^*, μ^* such that

$$\nabla_x L(X^*, \lambda^*, \mu^*) = 0$$

$$h_i(X^*) = 0, i = 1, 2, \dots, m$$

$$g_j(X^*) \leq 0, j = 1, 2, \dots, p$$

$$\mu_j^* g_j(X^*) = 0, j = 1, 2, \dots, p$$

$$\mu_j^* \geq 0, j = 1, 2, \dots, p$$

$$Y^T \nabla_{xx}^2 L(X^*, \lambda^*, \mu^*) Y > 0 \text{ on } M = \{Y: \nabla h(X^*)^T Y = 0, \nabla g_j(X^*)^T Y = 0 \text{ for all } j \in J\},$$

$$\text{where } J = \{j: g_j(X^*) = 0, \mu_j^* > 0\}.$$

Here, X^* is assumed to be *regular*, that is, $\nabla h_i(X^*), \nabla g_j(X^*), j \in J$ are linearly independent.



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Example:

$$\text{Min: } f(x) = (x_1 - 1)^2 + x_2 - 2$$

$$\text{Subject to: } h(x) = -x_1 + x_2 - 1 = 0$$

$$g(x) = x_1 + x_2 - 2 \leq 0$$

We have

$$\nabla h(x) = [-1 \ 1]^T, \nabla g(x) = [1 \ 1]^T$$

Thus, $\nabla h(x)$ and $\nabla g(x)$ are linearly independent and hence all feasible points are regular.

The Lagrange function is

$$\begin{aligned} L &= f(x) + (\lambda h + \mu g) \\ &= (x_1 - 1)^2 + x_2 + \lambda(-x_1 + x_2 - 1) + \mu(x_1 + x_2 - 2) \end{aligned}$$

The K-T conditions are as follows:



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$$\frac{\partial L}{\partial x_1} = 2x_1 - 2 - \lambda + \mu = 0 \quad (1)$$

$$\frac{\partial L}{\partial x_2} = 1 + \lambda + \mu = 0 \quad (2)$$

$$h(x) = -x_1 + x_2 - 1 = 0 \quad (3)$$

$$g(x) = x_1 + x_2 - 2 \leq 0 \quad (4)$$

$$\mu g = \mu(x_1 + x_2 - 2) = 0 \quad (5)$$

$$\mu \geq 0 \quad (6)$$

To find points that satisfy the above conditions, first try $\mu > 0$.

From eqns. (1), (2), (3), (5), we get

$$2x_1 - 2 - \lambda + \mu = 0$$

$$1 + \lambda + \mu = 0$$

$$-x_1 + x_2 - 1 = 0$$

$$x_1 + x_2 - 2 = 0$$



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Solving the above equations we obtain

$$x_1 = 0.5, x_2 = 1.5, \lambda = -1, \mu = 0$$

However, the above is not a legitimate solution to the K-T condition as we obtained $\mu = 0$, which contradicts the assumption that $\mu > 0$.

Next, try $\mu = 0$. Thus, we have

$$2x_1 - 2 - \lambda = 0$$

$$1 + \lambda = 0$$

$$-x_1 + x_2 - 1 = 0$$

and the solutions must satisfy eqn. (4). Solving the above equations gives

$$x_1 = 0.5, x_2 = 1.5, \lambda = -1$$

Note that $X = [0.5, 1.5]^T$ satisfies eqn. (4). Thus, the point X satisfies the K-T necessary conditions. Is it a minimizer?



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Note that as $\mu = 0$, the active constraint $g(X) = 0$ does not enter the computation of $\{Y: \nabla h(X^*)^T Y = 0, \nabla g_j(X^*)^T Y = 0 \text{ for all } j \in J\}$.

We have

$$\nabla h^T Y = [1 \quad 1] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = y_1 + y_2 = 0$$

$$\nabla_{xx}^2 L(X, \lambda, \mu) = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \geq 0 \text{ on } \mathbb{R}^2$$

On M'' ,

$$\begin{aligned} Y^T \nabla_{xx}^2 L Y &= [y_1 \quad y_2] \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \\ &= [y_1 \quad y_2] \begin{bmatrix} 2y_1 \\ 0 \end{bmatrix} \\ &= 2y_1^2 > 0 \text{ for all } Y \neq 0. \end{aligned}$$

This implies that $\nabla_{xx}^2 L$ is positive definite on M'' .

Thus, $X = (0.5, 1.5)^T$ is a minimum point.



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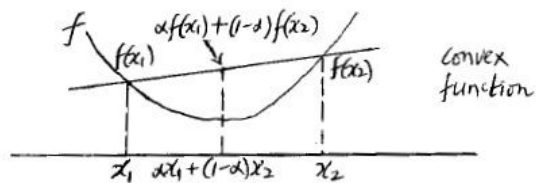
- **Convex programming problem**

In the previous section, we have seen that not all K-T points are optimal points. To guarantee that the solution is optimal, we need certain additional *convexity* assumptions.

Convex functions

A function $f(X)$ is defined as a convex function if for any two points X_1 and X_2 in the search space and for $0 \leq \alpha \leq 1$

$$f(\alpha X_1 + (1 - \alpha)X_2) \leq \alpha f(X_1) + (1 - \alpha)f(X_2)$$



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Geometrically it means that if a function is convex and if a line is drawn between any two points on the surface of that function, the line segment joining these two points will lie entirely above that function.

Note: A linear function is also convex.

A function f on a convex set is *concave* if $-f$ is convex.

The convexity (concavity) of a function is tested by checking the Hessian matrix of the function. If the Hessian matrix is positive definite or positive-semi-definite (*negative semi-definite*) for all values of X in the search space, the function is a convex (*concave*) function.



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Example

Determine whether the following functions are convex, concave or neither.

1. $f_1(x_1, x_2) = x_1^2 + x_2^2$
2. $f_2(x_1, x_2) = 2x_1x_2 - x_1^2 - x_2^2$

The Hessians of f_1 and f_2 are

$$H_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} > 0, H_2 = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} < 0$$

Hence, f_1 is convex over $\mathbb{R}^2$ and f_2 is concave over $\mathbb{R}^2$.



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K-T sufficient theorem

Let $f(X)$ be convex, the inequality constraints $g_i(X)$ be all convex functions and equality constraints $h_i(X)$ be linear. If there exists a solution (X^*, λ^*, μ^*) that satisfies the K-T conditions, then X^* is an optimal solution to the NLP problem.

Example

Consider the following NLP:

$$\text{Min: } f(X) = x_1^2 - x_2$$

$$\text{Subject to: } x_1 + x_2 = 6$$

$$x_1 \geq 1$$

$$x_1^2 + x_2^2 \leq 26$$



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We have

$$H(X) = x_1 + x_2 - 6 = 0$$

$$g_1(X) = -x_1 + 1 \leq 0$$

$$g_2(X) = x_1^2 + x_2^2 - 26 \leq 0$$

The Lagrangian function is given by

$$L = x_1^2 - x_2 + \lambda(x_1 + x_2 - 6) + \mu_1(-x_1 + 1) + \mu_2(x_1^2 + x_2^2 - 26)$$

We have

$$\frac{\partial L}{\partial x_1} = 2x_1 + \lambda - \mu_1 + 2\mu_2x_1 = 0 \quad (1)$$

$$\frac{\partial L}{\partial x_2} = -1 + \lambda + 2\mu_2x_2 = 0 \quad (2)$$

$$h(X) = x_1 + x_2 - 6 = 0 \quad (3)$$

$$g_1(X) = -x_1 + 1 \leq 0 \quad (4)$$

$$g_2(X) = x_1^2 + x_2^2 - 26 \leq 0 \quad (5)$$



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$$\mu_1 g_1 = \mu_1(-x_1 + 1) = 0 \quad (6)$$

$$\mu_2 g_2 = \mu_2(x_1^2 + x_2^2 - 26) = 0 \quad (7)$$

$$\mu_1 \geq 0, \mu_2 \geq 0 \quad (8)$$

We have to solve equations (1) – (8).

If $\mu_1 > 0, \mu_2 > 0$, then from (6) and (3), we obtain

$$x_1 = 1, x_2 = 5$$

From (1) and (2), we get

$$\mu_1 + 8\mu_2 - 3 = 0 \quad (9)$$

If $\lambda = 0$, then from (2),

$$\mu_2 = 0.1$$

And from (9),

$$\mu_1 = 2.2$$



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Thus, $X = (1, 5)^T$, $\mu = (2.2, 0.1)^T$, $\lambda = 0$, satisfies the K-T conditions.

Is X the minimum point?

The objective function has the following Hessian matrix:

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

We have $H_1 = 2$ and $H_2 = 0$. Thus, H is positive semi-definite and f is a convex function. Also, h and g_1 are linear functions. For g_2 , we

$$\text{have } \nabla^2 g_2 = \begin{bmatrix} \frac{\partial^2 g}{\partial x_1^2} & \frac{\partial^2 g}{\partial x_1 \partial x_2} \\ \frac{\partial^2 g}{\partial x_2 \partial x_1} & \frac{\partial^2 g}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Thus, $\nabla^2 g_2$ is positive definite. Both g_1 and g_2 are convex functions. We conclude that $X = (1, 5)^T$ is the minimum point.

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Sensitivity analysis

Assuming that an optimal solution is found, how does the optimal value of the objective function change if some parameters are changed marginally?

One way to know the solution is to simulate the revised optimal problem again, but this method may be computationally expensive.

However, a quick estimate of the solution can be achieved by using the optimal Lagrange multipliers corresponding to the constraint. This study of the effect of the parameters on the optimal objective function is known as sensitivity analysis. This analysis is usually performed after the optimal solution of the original problem has been found.

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Consider the NLP as follows:

Minimize: $f(X)$

Subject to: $h(X) = AX - B = 0$

We assume that f is twice continuously differentiable and that A is an $m \times n$ matrix of full row rank. The Lagrangian function is of the form

$$L_\lambda(X, \lambda) = f(X) + \lambda^T(AX - B)$$

We also assume that a local minimum X^* of f has been found, with a corresponding optimal objective value of $f(X^*)$.

Suppose now that the right-hand side B is perturbed to $B + \Delta B$, where ΔB is a vector of “small” perturbations. We shall investigate how the optimal objective value changes as a result of these perturbations. If the perturbations are *sufficiently small*, it is reasonable to assume that the new problem has an optimal



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that is close to X^* . For $\bar{X}$ close to X^* , with $A\bar{X} = B + \Delta B$, we can use a Taylor series approximation to obtain

$$f(\bar{X}) \approx f(X^*) - \sum_{i=1}^m \Delta B_i \lambda_i^*$$

Or

$$\Delta f(X^*) = f(\bar{X}) - f(X^*) \approx - \sum_{i=1}^m \Delta B_i \lambda_i^*$$

In particular, this is valid if $\bar{X}$ is the minimum of the perturbed problem. If the right-hand side of the i -th constraint changes by ΔB_i then the optimal objective value changes by approximately $-\Delta B_i \lambda_i^*$. Hence, λ_i^* represents the rate of optimal cost decrease as the level of the i -th constraint increases.



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Consider the NLP as follows:

Minimize: $f(X)$

Subject to: $h_i(X) = 0, i = 1, \dots, m$

$g_j(X) \leq 0, j = 1, \dots, p$

Let the Lagrangian function be

$$L(X, \lambda, \mu) = f(X) + \sum_{i=1}^m \lambda_i h_i(X) + \sum_{j=1}^p \mu_j g_j(X)$$

The above NLP can be rewritten as

Minimize: $f(X)$

Subject to: $h'_i(X) = \alpha_i, i = 1, \dots, m$

$g'_j(X) \leq \beta_j, j = 1, \dots, p$

where $h_i(X) = h'_i(X) - \alpha_i, g_j(X) = g'_j(X) - \beta_j$.

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Let

$$\Delta\alpha_i = (\alpha_i)_{\text{new}} - (\alpha_i)_{\text{old}}$$

$$\Delta\beta_j = (\beta_j)_{\text{new}} - (\beta_j)_{\text{old}}$$

Then the net change in the optimal objective function value is given by

$$\Delta f^* = f_{\text{new}} - f^* \approx - \sum_{i=1}^m \lambda^*_i \Delta\alpha_i - \sum_{j=1}^p \mu^*_j \Delta\beta_j$$

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Example:

Consider the following problem:

$$\begin{aligned} \text{Min } f(x) &= x_1^2 - 2x_1 + x_2^2 - x_3^2 + 4x_3 \\ \text{s.t. } x_1 - x_2 + 2x_3 &= 2 \end{aligned}$$

It can be determined that the optimal solution is given by

$$x^* = (2.5, -1.5, -1)^T, \text{ with } f(x^*) = -1.5 \text{ and } \lambda^* = -3.$$

Consider now the perturbed problem:

$$\begin{aligned} \text{Min } f(x) &= x_1^2 - 2x_1 + x_2^2 - x_3^2 + 4x_3 \\ \text{s.t. } x_1 - x_2 + 2x_3 &= 2 + \Delta \\ \text{and denote its min value by } f^*(\Delta). \end{aligned}$$



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We have

$$\Delta = (\alpha)_{\text{new}} - (\alpha)_{\text{old}} = (2 + \Delta) - 2$$

and

$$\begin{aligned} \Delta f^* &\approx -\lambda^* \Delta = -(-3)\Delta = 3\Delta \\ \Rightarrow f^*(\Delta) &\approx (f^*)_{\text{old}} - \lambda^* \Delta \\ &= -1.5 + 3\Delta \end{aligned}$$

If $\Delta = 0.5$, the approximate optimal objective value is 0.

The precise solution to the perturbed problem is $x_1 = 2.5 - \Delta/2$, $x_2 = -1.5 + \Delta/2$, and $x_3 = -1 + \Delta$ with an objective value of

$$f^*(\Delta) = -1.5 + 3\Delta - 0.5\Delta^2$$

If $\Delta = 0.5$, the true value of the optimal objective is -0.125.



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